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STANDARDS

Intelligent O-RAN Beyond 5G: Architecture, Use Cases, Challenges, and Opportunities

SIMONA MARINOVA AND ALBERTO LEON-GARCIA, (Life Fellow, IEEE)
Department of Electrical and Computer Engineering, University of Toronto, Toronto, ON M5S 3G9, Canada
Corresponding author: Simona Marinova (simona.marinova@utoronto.ca)

ABSTRACT Open RAN (Radio Access Network) is revolutionizing the telecom space by introducing a framework based on the concepts of virtualization and openness. O-RAN fosters virtualized and disaggregated RAN components connected via open interfaces based on specifications by the O-RAN Alliance. The network is optimized using RAN intelligent controllers (RICs), which can take data-driven, closed-loop actions in a RAN built in a multi-vendor, interoperable environment. The goal of this paper is to provide insights and guidance about the paradigm shift brought by O-RAN in order to create open, softwarized, intelligent and optimized networks. We focus on the intelligence aspects by providing an in-depth view of the near-RT and non-RT RICs specified by the O-RAN Alliance, including the architecture and interfaces. A novel aspect of this paper is that we provide guidelines in terms of the artificial intelligence and machine learning (AI/ML) approaches and frameworks that are useful in the O-RAN context, and consider the applications (xApps and rApps) that can be created to programmatically and autonomously control and optimize the network through the RICs for V2X, Industry 5.0, and other very demanding service types. Additionally, we provide the E2E network slice orchestration architecture, and demonstrate the suitability of O-RAN for the requirements of the service types to be achieved. Finally, we discuss research challenges and opportunities and overview existing experimental research platforms that are used to innovate and drive advances in the O-RAN effort.

INDEX TERMS AI/ML, B5G/6G, open RAN, O-RAN alliance, radio intelligent controller (RIC).

I. INTRODUCTION

The fifth generation (5G) networks introduce services and use cases with very aggressive demands in terms of throughput, latency, reliability, security, and privacy. These demands are met by 5G wireless systems that encompass new technologies such as network slicing [1], [2], [3], massive Multiple Input Multiple Output (MIMO) [4], millimeter wave and sub-terahertz communications [5], [6], signal processing aided by machine learning (ML) [7], [8], and ML-enabled network planning/optimization/diagnostics [9], [10], [11], [12]. These technologies introduce complexity that the management and orchestration systems built by the operators need to address.

Current RAN components are monolithic black boxes from the perspective of the operators, implementing the entire protocol stack on "specialized" hardware that is completely enclosed and vendor-specific. This approach is challenged by the business requirements for new applications such as vehicle-to-everything (V2X) and Industry 4.0 which require: 1) agile development and introduction of new features and capabilities; 2) complete observability of the RAN systems and ability to optimize performance; 3) full reconfigurability of the RAN Nodes and fine-tuning that addresses all the diversity in deployment and traffic forecasting; 4) coordination among different network nodes, in closed-loop manner, and at time scales suitable for applications such as V2X and Industry 4.0; 5) vendor openness and full interoperability, allowing vendors to choose RAN components to fulfill specific communication and business requirements of 5G and B5G networks and of slice types such as V2X and Industry 4.0.

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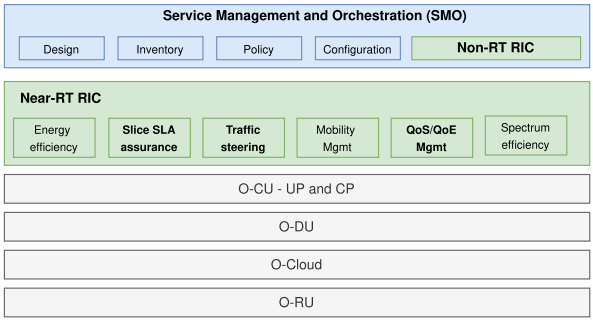


FIGURE 1. O-RAN architecture and components [13].

To achieve the business and communication requirements, cellular networks are undergoing a paradigm shift led by concepts such as softwarization, disaggregation, open interfaces, and generic programmable hardware to replace traditional specialized hardware boxes and architectures [14], [15], [16]. This shift paves the way for more *flexible, agile, cloud-native, interoperable, data-driven and ML-optimized networks* [17]. This new architectural design is under development in Radio Access Networks, primarily led by the Open RAN (O-RAN) Alliance. The O-RAN approach and architecture is founded on three pillars: 1) ***interface openness and disaggregation*** to support interoperability in multi-vendor environments where real-time monitoring and data collection, and interactions with the RAN are available; and 2) ***intelligence and closed-loop automation***, to support RAN optimization using ML-based closed-loop decision-making; and 3) *softwarized and cloud-native RAN* and the ability to leverage the flexibility and scalability of hyperscalers for future network service types. More specifically, O-RAN supports a RAN split that disaggregates the base station functionalities into a Radio Unit (RU), Distributed Unit (DU), and Centralized Unit (CU). These components are connected to RAN intelligent controllers through *open interfaces* from which telemetry is streamed from the RAN and control actions can be pushed down, as shown in Figure 1.

The O-RAN architecture includes two distinct ***RAN intelligent controllers (RICs)*** hosting ***applications*** that implement *policies and control actions* in two different time frames: 1) Near-RT RIC for control loops in 10 ms to 1s time range; 2) Non-RT RIC for control loops in more than 1 s. Control loops below 10 ms are currently investigated by the research community, e.g., work by D'oro et al. [18] and Ko et al. [19]. Some of the most notable applications and use cases [20], [21], [22] (a combination of applications) implemented on top of the RICs include energy efficiency, SLA slice assurance, QoS management, as seen in Figure 1. The implementation of these use cases relies on the O-RAN pillars of openness, disaggregation, intelligence, and cloud-native automation, and is what drives the evolving O-RAN architecture. We examine the most prominent use cases in the later sections.

TABLE 1. List of acronyms.

5G	Fifth generation
3GPP	Third generation partnership project
5GPPP	5G infrastructure public private partnership
AI	Artificial intelligence
B5G	Beyond 5G
CaaS	Container as a service
CAPEX	Capital expenditures
CM	Configuration management
CP	Control plane
CSMF	Communication service management function
CU	Centralized unit
DAF	Data analytics function
DU	Distributed unit
EI	Enrichment information
E2E	End-to-end
ETSI	European telecommunication standards institute
FL	Federated learning
FM	Fault management
gRPC	Google remote procedure call
IETF	Internet engineering task force
KPI	Key performance indicator
KPM	Key performance measurement
MAC	Medium access control
MANO	Management and orchestration
MIMO	Multiple input multiple output
NFV	Network function virtualization
NSMF	Network slice management function
NSSMF	Network slice subnet management function
NSSI	Network slice subnet instance
NWDAF	Network data analytics function
O-Cloud	Open cloud
ONAP	Open network automation platform
OPEX	Operational expenditures
O-RAN	Open radio access network
OSC	Open source community
OSM	Open source management and orchestration
OSS/BSS	Operations support system/Business support system
PCA	Principal component analysis
PDU	Packet data unit
PM	Performance measurement
PRB	Physical resource block
QoE	Quality of experience
QoS	Quality of service
RAT	Radio access technology
RC	Radio Control
RIC	Radio intelligent controller
RL	Reinforcement learning
RRC	Radio resource control
RT	Real-time
RU	Radio unit
SBA	Service-based architecture
SDK	Software development kit
SLA	Service level agreement
SMO	Service management and orchestration
UAV	Unmanned aerial vehicle
UP	User plane
V2X	Vehicle to everything
WG	Work group

A. CONTRIBUTIONS

The O-RAN paradigm marks a massive change in how we build and operate networks. O-RAN networks will enable the introduction of transformative applications in the Near-RT and Non-RT RIC based on ML approaches. In this paper, we examine the following:

- O-RAN components and specifications centered around the non-RT/near-RT RICs;

- AI/ML workflows and closed-loop control automation;
- Use cases such as RAN SLA assurance, traffic steering, QoS/QoE for next generation applications; both from the standards and research perspectives;
- Research challenges and opportunities for the development of the O-RAN RIC architecture and the AI/ML foundation;
- Finally, a comprehensive view of the current large-scale research testbeds supporting O-RAN RIC and use case R&D.

The paper is organized as follows. Section II provides an overview of the key architectural principles, culminating in a high-level introduction of the O-RAN architecture. Sections III and IV provide an introduction to the Near-RT and Non-RT RIC, respectively, as we begin each section with a discussion on the desired outcome and progressively delve into detailed description of features and functionalities for the RICs. In Section V we consider the appropriate ML approaches and implementations to address the ORAN-specific use cases. Section VI covers the ORAN-specific discussions by providing an overview of large-scale O-RAN research platforms. In Section VII, we provide insights into the E2E network slice orchestration framework components and requirements, encompassing the O-RAN architecture. In Section VIII, having considered the end-to-end network slice view and all the components necessary from an operator perspective, we delve into O-RAN use cases. Section IX identifies research challenges that need more investigation. For the sake of completeness and to provide more context to enable the readers to dig into the O-RAN standardization activities, in Section X we present an overview of the work groups, focus groups, and research groups. Finally, Section XI concludes the paper.

II. KEY ARCHITECTURAL PRINCIPLES

O-RAN is a disruptive technology in that it aims to open up the interfaces between software-based components in order to provide a number of benefits such as reduced costs, dynamic lifecycle of services, faster time to market, and stimulating innovations in the field. In this section, we focus on the foundational principles of O-RAN and the resulting O-RAN components.

A. VIRTUALIZATION AND SOFTWARIZATION

The first principle of the O-RAN architecture is virtualization and softwarization of the components and their deployment on a cloud computing platform called O-Cloud. This platform is a set of computing resources and a virtualized infrastructure usually spanning across different geo-distributed datacenters. The O-Cloud includes the management and orchestration functionalities [23] for O-RAN, and focuses on optimization of the RAN functionalities from the underlying infrastructure to the virtualization technology that hosts the network functions [24].

Multiple workgroups (WGs) within the O-RAN Alliance are providing guidance on O-Cloud from different perspectives. For example, in WG6 O-RAN is developing standardized hardware acceleration abstractions called Acceleration Abstraction Layers (AALs) that define APIs between dedicated hardware-based logical processors and the O-RAN softwarized infrastructure. In WG7 there is a focus on the characteristics and requirements of the general-purpose hardware that implements the O-RAN nodes (CU, DU, RU).

B. O-RAN DISAGGREGATION AND OPEN INTERFACES

By focusing on the openness and standardization of interfaces in the RAN architecture, O-RAN aims to provide an alternative to vendor lock-in solutions and an opportunity for network function selection solely based on performance. Functional splits in the RAN have been introduced since 3GPP Release 15 in order to disaggregate the base station functionalities into a Centralized Unit (CU) and Distributed Unit (DU) that implement the higher and lower layers of the RAN protocol stack, respectively. To provide flexibility in standardization, 3GPP has introduced a number of functional splits of the protocol stack. Prior to O-RAN, the implementations of these functional splits were proprietary to the vendor and used vendor-defined interfaces. O-RAN has adopted solutions for the fronthaul interface between the RU and DU, and for the F1 interface between the DU and CU. These solutions use the enhanced common public radio interface (eCPRI), a modification to the proprietary CPRI interface.

C. INTELLIGENT DECISION-MAKING

Once the RAN protocol stack is flexibly implemented leveraging *softwarization* and *openness* principles, the next step is to provide intelligence for RAN control and optimization. The goal is for O-RAN to provide intelligence in Radio Intelligent Controllers (RICs) operating in different control loops, near real-time (10ms-1s) and non-real-time (above 1s), as well as control and optimization functionalities to the CU/DUs of different vendors.

The control loops in O-RAN applications can generally be deployed in two ways:

1) OPEN-LOOP CONTROL

Open-loop control is used when the applications are deployed as part of the RIC to provide *recommendations* where a domain expert needs to validate the action suggested by the application. An application may be collecting data and training an AI/ML model that could suggest an action to optimize the RAN for a certain objective. The action must be validated by a RAN expert who will execute it by hand. Open-loop control is frequently a first step towards closed-loop control.

2) CLOSED-LOOP CONTROL

Closed-loop control is the means for fully autonomous operation. In contrast to open-loop control, every step in the

loop is executed by the application and there is no human intervention.

D. CLOUD-NATIVE O-RAN

Cloud-RAN (C-RAN) was introduced prior to O-RAN and referred to the softwarization of RAN components and centralization of RAN resources in a cloud location in order to optimize the resource allocation for specific objective function, e.g., computing resource allocation, power allocation, etc.

O-RAN as a technology relies on softwarization of the components, as well as on openness, disaggregation, intelligence, and automation. *Cloud-native O-RAN* goes a step beyond and refers to *building and autonomously running the RAN components* in cloud platforms to take advantage of distributed computing [25]. Cloud native DU and CU are designed and built to exploit the scale, elasticity, resiliency, and flexibility the cloud provides, for example from hyperscalers like Google and Amazon.

E. O-RAN ARCHITECTURE

Figure 2 presents the resulting high-level view of the O-RAN components. The Radio Access Network (RAN) functionalities are implemented in a disaggregated manner between the open Radio Unit (O-RU), Distributed Unit (DU), and Centralized Unit (CU). O-RAN introduces the RAN Intelligent Controllers, Near-RT RIC and Non-RT RIC, which provide control loops in different time scales and serve different purposes in an operator network [26]. Finally, the Service Management and Orchestration (SMO) provides the end-to-end (E2E) orchestration capabilities [27].

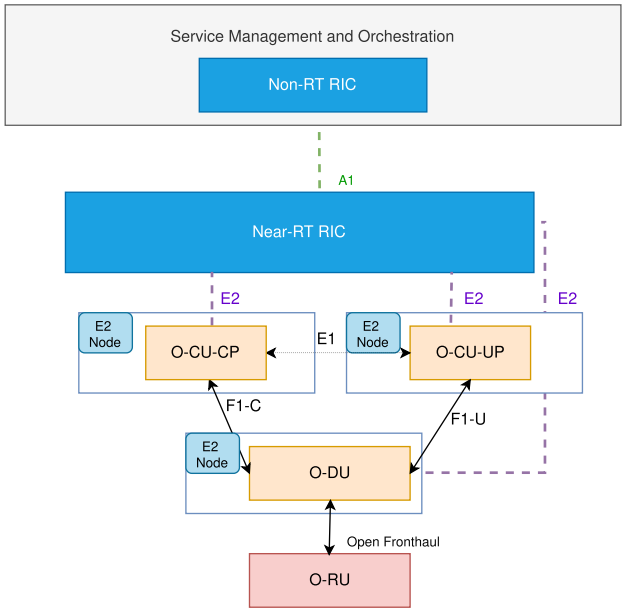


FIGURE 2. Near-RT RIC architecture and interfaces.

In the next Sections, we will provide more detailed description of the Near-RT, Non-RT RICs, and the ML frameworks that provide the tooling for intelligent decision-making.

III. NEAR-RT RIC

The Near Real Time Radio Intelligent Controller (Near-RT RIC) supports near real-time requirements and is a logical network node placed between the Service Management and Orchestration (SMO) layer and the RAN nodes (otherwise referred to as E2 Nodes) [28], as seen in Figure 2. The Near-RT RIC is connected to one Non Real Time Radio Intelligent Controller (Non-RT RIC) through the A1 interface. It is connected to multiple E2 Nodes: Open - Central Unit - Control Plane (O-CU-CP), Open - Central Unit - User Plane (O-CU-UP), and Open Distributed Unit (O-DU). Each of these E2 Nodes is connected to one Near-RT RIC at any given moment.

The Near-RT RIC can host one or more xApps (in its application layer) that use the E2 interface to collect near real-time information and determine output actions that maximize some metric [29]. Through the A1 interface, the Near-RT RIC can receive declarative policies and data enrichment information from the Non-RT RIC.

The E2 interface is a major contribution of the O-RAN architecture, which provides DU/CU direct communication to the near-RT RIC. This means that data types can be streamed from the E2 Nodes directly to the RIC platform providing the capability to collect data in seconds and less, as opposed to previous element management systems that would normally collect data every 15-30 mins. Furthermore, the E2 interface allows for the near-RT RIC applications (xApps) to reconfigure parameters in the E2 Nodes in order to control and optimize resources for a certain goal (e.g., energy efficiency, QoE per application, etc.) in near-RT (sub-1s).

The following subsections will introduce the E2 service models and then provide details of the important near-RT functions and requirements, as well as a deep dive into the E2 interface and capabilities.

E2 interface termination is used to: 1) terminate the E2 interface from E2 Nodes; 2) route xApp content to the target xApp; and 3) route non-xApp-related content to the E2 Manager (residing in the RIC). Finally, *Installed xApps* which are created in order to: 1) provide Radio Resource Management (RRM) control functionalities to be executed in the Near-RT RIC and enforced in the E2 Nodes through the E2 interface; 2) provide xApp-related messages to and from the E2 nodes through the E2 interface.

A. E2 SERVICE MODELS

The O-RAN Alliance has developed the following four E2 service models (E2SM) to provide a standardized and structured way to exchange information between the E2 Nodes and the Near-RT RIC through the E2 interface [30].

1) The ***E2SM Key Performance Measurements (KPM)*** service model is used to report performance measurements from the RAN [31].

2) The ***E2SM Network Interfaces (NI)*** service model is used for the transmission of messages from the E2 Node to the Near-RT RIC.

3) The ***E2SM RAN Control (RC)*** service model is responsible for the optimization of radio resource utilization in the E2 Nodes in a *closed-loop* manner [32]. It includes a multitude of control domains, including radio bearer control, radio resource allocation control, mobility control, radio access control, Carrier Aggregation (CA) control, and idle mobility control.

4) The ***E2SM Cell Configuration and Control (CCC)*** service model is responsible for reporting of cell-level and slice-level configuration in the RAN Nodes [33]. This service model provides slice-awareness to the RAN, which is very important to realizing dynamic network slicing and O-RAN WG1 use cases such as RAN slice SLA assurance.

B. NEAR-RT RIC REQUIREMENTS

The Near-RT RIC is expected to meet several *requirements*. First, by definition, it should support latency requirements for near-real-time optimization, *from 10 milliseconds up to 1 second*. Second, the architectural view is that the Near-RT RIC uses a dedicated *E2 connection* that uniquely identifies each E2 Node configured so as to directly connect and provide RIC-related information to the Near-RT RIC. Furthermore, the Near-RT RIC supports E2 connections from multiple E2 Nodes (that is, in a one-to-many architecture), and Radio Access Technology (RAT) agnostic and so supports multiple RAT types. Thus the workflow has the Near-RT RIC obtain from the E2 Nodes a list of functions supporting RIC-related Services and the corresponding E2 Service Model (E2SM). Third, the Near-RT RIC has the ability to host applications, known as xApps. Each individual xApp in the Near-RT RIC addresses specific RAN Functions in a specific E2 Node.

Fourth, the Near-RT RIC is designed for resiliency. For this reason, the O-RAN design allows E2 nodes to function independently of the Near-RT RIC when the E2 interface and/or Near-RT RIC fail. Fifth, the Near-RT RIC is connected to the Non-RT RIC through the A1 interface. Functionally, the A1 interface is used to provide Policies to the Near-RT RIC which can be used to modify the overall E2 Node behavior, namely the behavior of Near-RT RIC and its xApps. In addition, the Near-RT RIC is connected to the Service Management & Orchestration layer (through interface O1) for element management and configuration, shown later in Figure 5.

C. E2 INTERFACE - NEAR-RT RIC TO E2 NODES

A very important aspect of O-RAN is openness and the interface specifications that allows full reconfiguration of RAN nodes in an interoperable manner (i.e., without vendor lock-in). The E2 interface is specified to fulfill the openness aspect and to operate in sub-1 second, and thus enable closed-loop optimization of the extremely latency-sensitive RAN PHY and MAC parameters.

The relationship between the Near-RT RIC and the E2 Nodes is depicted in Figure 3. In addition to one or more xApps and the E2 termination, the Near-RT RIC also has a database storing data from the xApps and from the E2 Nodes. Each E2 Node has: 1) an E2 Agent that terminates the E2 interface and sends/receives E2 messages; 2) RAN functions that are controlled by the Near-RT RIC to support Near-RT RIC Services; and 3) other RAN functions that support other services.

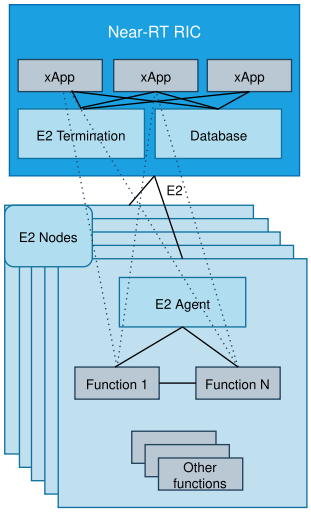


FIGURE 3. Near-RT RIC and RAN (E2) nodes.

The E2 interface can be used to collect data in near real-time as well as to push actions towards the E2 Nodes. The E2 interface must be open, and the exchange model between the E2 Nodes and the Near-RT RIC must be clearly defined and implemented. Thus the E2 interface supports the exchange of control signaling information between the E2 Nodes and Near-RT RIC through well-defined messages. Furthermore, the E2 interface specifies the action set that can be used to reconfigure the E2 Nodes when defined events occur. Conversely, the E2 interface enables the E2 Nodes to specify supported functionalities to the Near-RT RIC.

Simply put, E2 enables the Near-RT-RIC to: 1) receive predefined information (i.e., key performance measurements) from the E2 Nodes based on a predefined time scale (e.g., every 5 min) or proactively defined trigger event (e.g., significant change in measurements); and 2) push control actions and reconfigurations to the E2 Nodes.

D. E2 INTERFACE FUNCTIONS

We now delve into the specific functions of the E2 interface, which are grouped into: *Near-RT RIC Services* and *Near-RT RIC support functions*.

1) NEAR-RT RIC SERVICES

There are four *Near-RT RIC Services* as depicted in Figure 4.

The first service, ***E2 REPORT***, is composed of *E2 RIC Indication* messages containing data and telemetry from the E2 Node. To activate this service, an xApp in the Near-RT RIC subscribes to a function offered by the E2 Node. During the subscription, information specifying trigger events or the time scale at which the Node should send the report messages. The E2 Node sends reports accordingly through the E2 interface.

The second service is ***E2 INSERT*** which is composed of messages from the E2 Node to the xApp in the Near-RT RIC. An xApp subscribes to activate the service.

The third service, ***E2 CONTROL***, can be autonomously initiated by the Near-RT RIC or triggered after the reception of an *insert* message at the RIC. The service uses two messages: *RIC Control Request* from the RIC to the E2 Node; *RIC Control ACK* from the E2 Node to the RIC.

The last service, ***E2 POLICY***, specifies a subscription procedure involving: an event trigger; and a policy for the E2 Node to follow to perform Radio Resource Management (RRM).

2) NEAR-RT RIC SUPPORT FUNCTIONS

The Near-RT RIC support functions include the lifecycle procedures depicted in Figure 4.

The ***E2 Setup*** procedure establishes the E2 interface between the Near-RT RIC and an E2 Node. It involves the exchange of the following information: a list of supported Near-RT RIC services and the mapping of services to functions within the E2 Node (Figure 3); and a list of E2 Node configuration information.

The ***RIC Indication procedure*** transfers Report and/or Insert RIC Service actions associated with a RIC Subscription procedure and initiated by the E2 Node.

The ***E2 Reset*** procedure resets the E2 interface, from either the Near-RT RIC or an E2 Node. All previously exchanged information (i.e., E2 Setup, configuration updates, service updates, etc.) is kept after a reset.

The ***Near-RT RIC Service Update*** procedure is used by the E2 Node to notify the Near-RT RIC of changes to the list of supported services and the mapping of services to functions. It can be instantiated by a *RIC Service Query* message. An *E2 Node Configuration Update* is used by the E2 Node to notify the Near-RT RIC of configuration changes to the E2 Node or E2 connection.

E. REALIZATION OF THE E2 SERVICE MODELS

We complete the section by revisiting the E2 service models and discussing how they are realized using the E2 Interface and its functions.

The *E2SM KPM* reports performance measurements from the RAN using the E2 report service. During the E2 Setup procedure, the E2 Node advertises the metrics it can expose. An xApp hosted on the Near-RT RIC will then send a subscription message specifying the KPMs of interest, and whether the report should be generated periodically or trigger-based. The E2 Node uses *E2 Indication messages* of type *report* to stream the selected KPMs.

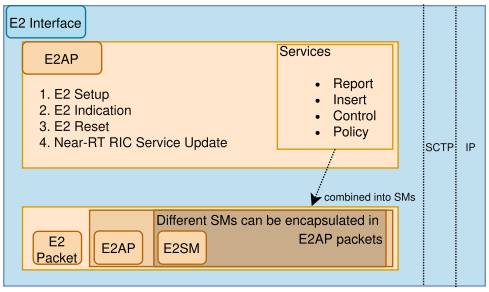


FIGURE 4. E2 interface specification: The E2 Application Protocol (E2AP) packet [34] includes the E2 Service Model (E2SM) payload. The E2 payload is encapsulated in SCTP and IP.

The *E2SM NI* transmits *report* messages from the E2 Node (received on specific network interfaces) to the Near-RT RIC domain through the E2 interface. The E2 Node is responsible for advertising which interfaces it supports during the subscription procedure.

The *E2SM RAN Control (RC)* handles the optimization of radio resource utilization in the E2 Nodes in a closed-loop manner.

The ***E2SM CCC*** service model is concerned with reporting of cell-level and slice-level configuration in the RAN Nodes. In terms of control services, it is responsible for slice resource allocation control, i.e., controlling slice specific resources in RAN Nodes such as slice specific uplink and downlink resource block allocation at the DU.

IV. NON-RT RIC

Non-RT RIC is the second component of the O-RAN RIC platform, and it is usually hosted alongside the Service Management and Orchestration (SMO) framework as shown in Figure 5. It provides a subset of the functionalities of the SMO framework and it supports applications that provide intelligent RAN control and optimizations in *timescales longer than 1s* [35]. The Non-RT RIC consists of two distinct components: the *Non-RT RIC framework* as defined by the O-RAN Alliance; and the *rApps* that are implemented on top of it [36]. The framework is responsible for the termination of the A1 interface from the Near-RT RIC, as well as the exposure of R1 services to the set of applications running on top of the framework.

The applications (rApps) leverage the existing functionalities in the Non-RT RIC and SMO (such as data collection and exposure, SDK for ML development, etc.) to provide additional intelligence and data-driven decision-making related to RAN control and optimization. The scope for the rApps includes radio resource management, data analytics, enrichment information for the xApps (hosted on the Near-RT RIC and operating in much more stringent control loops). Finally, the Non-RT RIC logically terminates the *A1 interface* which connects it to the Near-RT RIC, and

is used to provide policy-based management, Enrichment Information (EI), and ML model management functionalities.

A. R1 INTERFACE - RAPPS TO NON-RT RIC

The R1 interface is placed between the Non-RT RIC and the applications that reside on top of it (rApps) [37]. The interface allows rAPPs to query data and service availability, and to input policies that are to be implemented/relayed to the near-RT RIC and to E2 Nodes through the A1 and O2 interfaces, respectively.

The R1 services are produced by logical functions in the Non-RT RIC or by the rApps. As seen in Figure 5, the R1 termination enables the message exchange between the Non-RT RIC and the rApps.

1) SERVICE MANAGEMENT AND EXPOSURE SERVICES

The service management and exposure services can be accessed through the R1 interface and they include service registration and service discovery, authentication, authorization, and communication support services. In the context of service registration and discovery, a service producer is an entity that can send a registration request to register services.

2) DATA MANAGEMENT AND EXPOSURE SERVICES

The data management and exposure services are concerned with delivering the data which is created or collected by the producers and forwarded to consumers according to their needs. These functions are logical components that produce the data registration, data discovery, data subscription, data request, data delivery, and consume the data collection services.

An important aspect of these services is tracking the data types that are registered and matching data discovery requests against data types. If the data type is registered, then the functions identify a producer, in the Non-RT RIC/SMO (e.g., O1-related) or an rApp producer. These functions can also include the pre-processing of data such as quantization, normalization, correlation, labeling, etc.

When consumers subscribe to obtain data, the data management and exposure services check for the availability of that data. If available, the services then deliver the data. If not available, the services perform data collection from an identified data producer and then provide the data to the consumer. Optionally, the data collected from the producers can be stored in a data repository (with pre-processing before being stored or post-processing after being delivered) and read by the consumer.

3) A1-RELATED SERVICES

Since rApps can provide enrichment information and policy-related information to the Near-RT RIC, *A1 related services* are needed to provide access to functionalities related to the A1 interface. The logical functions that produce the A1 policy-related services are called *A1 policy functions*, and realize a number of functionalities.

A1 conflict mitigation is performed when several rApps generate policies with contradicting A1 policy statements. In this case, the A1 policy function has to decide whether to reject the generated policy or provide/request modifications before the policies are enforced.

A1 termination and Near-RT RIC interfacing includes functionalities that the A1 policy function needs to provide, including: 1) determining the Near-RT RIC for the generated policies; 2) sending the A1 policy creation, update, and deletion requests to the Near-RT RIC; 3) querying supported A1 policy types of the connected Near-RT RICs; 4) querying the status of the enforced policies; and 5) receiving the A1 policy feedback from the Near-RT RIC.

As mentioned above, A1 can also provide *Enrichment Information (EI)* for rApps related services via the R1 interface. As such, functionalities for the EI can include: 1) handling of EI types registered by rApps via the R1 interface; 2) handling of EI jobs created by the Near-RT RIC via the A1 interface; and 3) data collection delivered to the Near-RT RIC via the A1 interface.

4) O1-RELATED SERVICES

The O1-related services provide access to functionalities between the R1 and O1 interface, and produce O1-related services to rApps via the R1 interface. These services also include the *configuration management (CM) conflict mitigation*. The *interfacing with Near-RT RICs and E2 Nodes* is provided through the O1 termination (as seen in Figure 5). The O1-related functions for this functionality include: 1) reception of fault notifications and alarm lists from the Near-RT RICs and E2 Nodes; 2) provisioning of configuration changes to the Near-RT RIC and E2 Nodes; and 3) collection of performance measurement (PM) data from the Near-RT RIC and E2 Nodes.

5) O2-RELATED SERVICES

The O2-related services provide functionalities between the R1 and O2 logical interface within the architecture, and produce the O2-related services to rApps via the R1 interface. The relevant functionalities include: 1) production of O2-related services to rApps over the R1 interface; 2) interfacing with O-Cloud through the O2 termination (as seen in Figure 5) in order to receive the O-Cloud infrastructure and deployment performance, configuration, and fault management data, as well as the configuration changes; and 3) conflict mitigation for configuration changes.

B. A1 INTERFACE—NEAR-RT RIC TO NON-RT RIC

The O-RAN architecture includes the A1 interface, to connect the Near-RT and Non-RT RICs [38], [39]. Normally the Non-RT RIC is implemented within the SMO layer, so that the Non-RT RIC entity handles deployment, configuration, and data collection of the RAN nodes. The SMO layer is also responsible for the implementation of the artificial intelligence (AI) and ML workflow and coordination of the

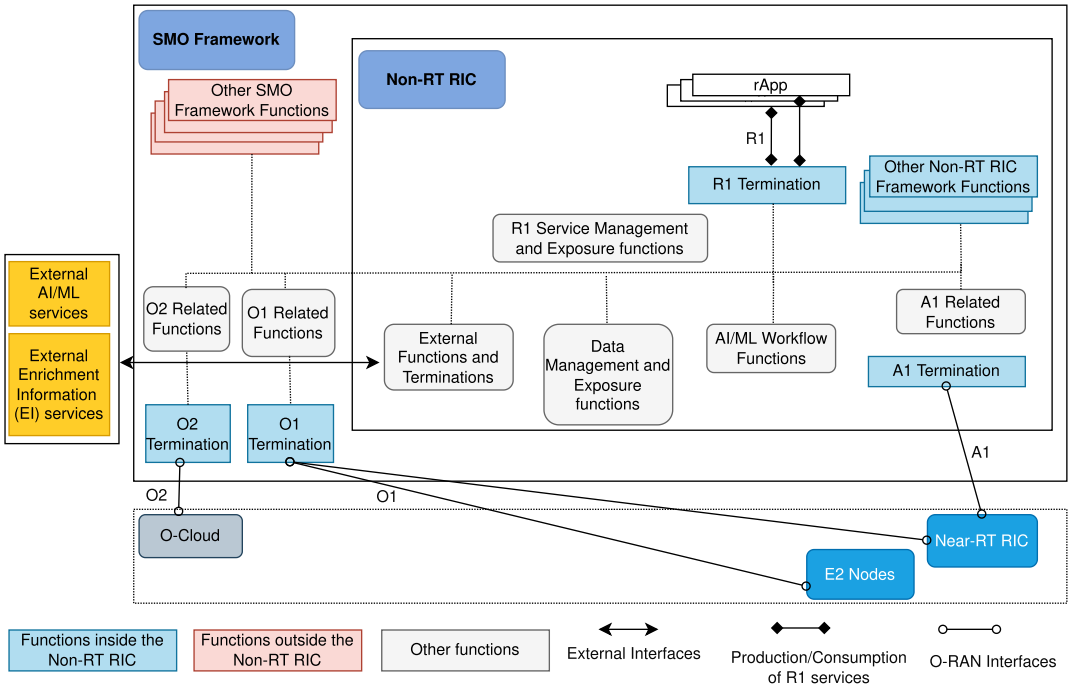


FIGURE 5. Non-RT RIC generic architecture and components.

training, validation, testing of different ML approaches. The SMO layer also has access to relevant data other than the one collected from the RAN nodes; this is called Enrichment Information (EI) and can be used to enhance the RAN optimization efforts.

The A1 interface plays a central role in one of the driving use cases in O-RAN, namely, the optimization of the RAN performance to satisfy SLAs. The A1 interface enables the Non-RT RIC to provide policy-based guidance through ML model management and enrichment information that it supplies to the Near-RT RIC. The RRM can then perform optimally even under varying conditions.

1) A1 INTERFACE GENERAL PRINCIPLES

The A1 interface general principles guide the creation and evolution of the A1 interface. As with E2, the most important capability is the *openness of the interface between the Non-RT and Near-RT RIC*. This openness enables multi-vendor deployments by making the interface independent of Non-RT and Near-RT RIC implementations from different vendors. Furthermore, to support extensibility additional capabilities can be added in terms of service and data type support without requiring protocol/procedure changes.

Another A1 interface consideration is the *policy-based guidance* of the RRM functions and xApps in the Near-RT RIC. Policy feedback is carried from the Near- to Non-RT RIC, while policy enforcement service is provided in the Near-RT RIC. Policies transferred through the A1 interface follow a standard syntax and language.

2) ENRICHMENT INFORMATION (EI)

EI is needed to provide visibility into application-level performance, for example in terms of throughput, latency, jitter, frame loss, etc. EI can be imported directly from applications, from the management components in the 5GC, or even from social media (e.g., news about upcoming events that can influence the resource allocation). There are currently no standardized components for EI nor are there standardized application-specific KPIs; each application can be customized in terms of which KPIs to track and extract.

In the O-RAN architecture, the EI is transferred from the Non-RT RIC to the Near-RT RIC through the A1 interface, and A1-policy lifecycle management is provided by the Non-RT RIC. For example, if the RAN indicates the fulfillment of a RAN intent over the O1 interface, it is the responsibility of the Non-RT RIC to delete the A1 policy on that specific intent. The lifecycle management of EI jobs is done by the Near-RT RIC according to the internal functions and the xApp requirements. EI is delivered by the Non-RT RIC according to the EI job created by the Near-RT RIC, until the job is deleted.

3) A1 APPLICATION PROTOCOL (A1AP)

The A1AP consists of APIs for the services defined in the A1 interface as seen in Figure 6: policy management, enrichment information, and ML model management.

4) A1 POLICY MANAGEMENT SERVICE

The A1 policy management service is used by the Non-RT RIC for the creation and management of A1 policies

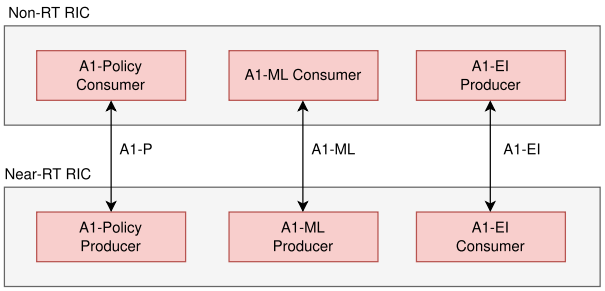


FIGURE 6. A1 services offered between the Near-RT and Non-RT RIC.

that are pushed down and enforced by the Near-RT RIC. The Non-RT RIC is the entity responsible for the lifecycle management of the policies, which includes the following functions: 1) create, update, delete the A1 policies; 2) query the status of the A1 policy; and 3) incorporation of A1 policy feedback from the Near-RT RIC, which is coupled with observability from the O1 interface to determine whether the RAN intent is fulfilled.

5) A1 ENRICHMENT INFORMATION SERVICE

As mentioned in the general principles above, the EI provides valuable information (which is not available in the RAN nodes) to the Near-RT RIC to improve the RAN optimization performance. This information can be internal or external to the O-RAN framework, and the A1 interface is used to send the EI from the Non-RT to the Near-RT RIC. There are different EI types, identified with a field EiTypeId, and are subject to different properties and delivery specifications through the A1 interface. The SMO is responsible for the setup of external and internal sources for EI. The Non-RT RIC is responsible for the discovery of the EI types over the A1 interface.

The Near-RT RIC is responsible for the management of the lifecycle of EI jobs, from creation when EI of specific type is required, to the deletion when the Near-RT RIC no longer needs that information. After a job is created, the Near-RT RIC is responsible for querying EI.

6) A1 ML MODEL MANAGEMENT SERVICE

For completeness, we include this interface service as part of this section. however, at the moment of writing there are no specifications about this planned service.

C. O1 INTERFACE—SMO TO O-RAN MANAGED ENTITIES

The O1 interface connects the O-RAN E2 Nodes to the Non-RT RIC in similar fashion to how the Near-RT RIC connects to E2 Nodes. This interface provides extended *operations and maintenance* capabilities, including the lifecycle management of the components, performance assurance, data collection and KPI reports, software and collection management [40].

1) PERFORMANCE MANAGEMENT AND ASSURANCE

Performance management and assurance in O1 is used to stream in real time or to report through file transfer the performance data to the SMO/Non-RT RIC, for use in analytics and AI/ML, similar to how the E2SM KPM is used in the E2 interface. The SMO selects the KPIs required and the reporting frequency [41]. The KPI reporting use cases and formats are based on existing 3GPP specifications [42], [43], [44], [45], [46], and on the VNF event stream (VES) project. The performance metrics can be 3GPP-based, vendor-specific, or O-RAN standardized. For non-RT bulk reports, a file-ready notification from the E2 Nodes is sent to the SMO via HTTP APIs. The file transfer itself is carried over SFTP. On the other hand, for streaming in real time, WebSockets are created. The SMO can also monitor trace-based events, e.g., RRC connection established, radio link failures, etc.

The O1 interface is used to push configurations and download files on the nodes managed by the SMO, similar to how the E2SM RC is used to provide radio control in a near-rt fashion. For instance, SMO carries out software updates for the O-DU/O-CU, configuration files for the O-RU, deployment of ML models in real-time, security configurations and certifications, etc.

2) PROVISIONING MANAGEMENT

The provisioning management functionality is used by the SMO to push configurations to the E2 Nodes, and by E2 nodes to report their configurations to the SMO. The O1 interface uses REST/HTTPS APIs and NETCONF [47], a protocol standardized by the Internet Engineering Task Force (IETF) for the lifecycle management of the managed entities in the network. The provisioning type support is defined by 3GPP [23].

3) FAULT MANAGEMENT

The fault management functionality is concerned with event- and error- reporting by E2 Nodes to the SMO. It is based on 3GPP specifications [23], [44] and the E2 Nodes report errors in a JSON format through REST APIs. On the other hand, the SMO can query information about alarms, and decide how to act, that is, clear the alarm, change configuration, etc.

A basic, but very valuable capability for visibility is *heartbeat management*, which is used to provision heartbeats on the E2 Nodes to monitor the status and availability of the Nodes and the services running on them.

D. O2 INTERFACE—SMO TO O-CLOUD

The O2 interface is used to connect the SMO to the O-RAN O-Cloud, mainly for management and provisioning of network functions in an automated manner [48]. The goal is to provide visibility of the inventory of E2 Nodes controlled by the O-Cloud, as well as monitoring, provisioning, fault tolerance, performance management.

The O-RAN Alliance is working towards the standardization of the O2 interface and its required capabilities, as well as consolidating and aligning the work with standards bodies and open-source projects that have intersecting features and capabilities to those required by the O2 interface and its associated entities. ETSI and 3GPP are the most relevant standards bodies, and the open-source projects include OpenStack and Kubernetes as the cloud platforms, as well as ONAP [49] and OSM [50] as the ETSI MANO and Orchestration tools.

V. AI/ML WORKFLOWS

In this Section, we will provide an overview of the ML approaches and models for O-RAN use cases, as well as the placement of ML pipelines in the O-RAN architecture. Novel slice service types such as V2X, gaming, and Industry 4.0 pose more stringent communication requirements, and the spatio-temporal fluctuations in user demands require that the operators promptly and proactively allocate resources and reconfigure the networks to use resources efficiently and save on CAPEX and OPEX. As the need for data quality and quantity increases, AI/ML pipelines and AI/ML-based decision-making is becoming more prevalent to achieve automated, data-driven control, planning, and optimization of resources in the network.

In Sections III and IV, we presented the architecture and components of the Near-RT and Non-RT RIC, which could be viewed as providing the abstraction layer for different xApps and rApps to control and optimize the network. RAN data types are sent to the RICs through several interfaces, and different types of applications (e.g., energy efficiency, QoS resource allocation, etc.) can subscribe to receive different data types with different frequency (e.g., every min, every hour, etc.). The applications then use the data as input to the ML framework to decide whether an action is required to be able to achieve the objectives or desired behaviour (e.g., min energy consumption, guaranteed QoS per user, etc.).

A. GENERAL PRINCIPLES AND REQUIREMENTS

Generally speaking, ML in production networks has two main phases: 1) Learning, during which an ML model is developed based on an appropriate dataset; 2) Inference, during which the ML model is deployed online to process collected data and perform classification or regression.

ML models can operate at a variety of time scales, and so several configurations are anticipated in the context of O-RAN. The Non-RT RIC is part of the SMO and normally where training is performed, as shown in Figure 7. The trained model is deployed for inference in either the Non-RT RIC or the Near-RT RIC depending on the control loop latency expectations, hardware requirements, and interface through which data collection is performed.

1) OFFLINE LEARNING

Offline learning is normally preferred (including reinforcement learning which is usually performed online) due to the nature of the network environment, which is prone to misconfiguration and errors leading to outages. The proposed approach is for the model to first be trained with offline data and that the trained model then be deployed in the network for inference.

2) TRAINING, TESTING, AND VALIDATION

These steps take place before the model is deployed in the network. This is especially important when considering the closed-loop scenarios required for full network automation. Unlike other fields such as robotics, untrained models are not deployed in the network.

3) ML APPLICATION MODULARITY

Modularity refers to the requirement that the ML applications be deployed fully decoupled from the data, that is, the ML application does not need to be aware where the data is coming from, whether it is from an interface like E2 or O1, or another ML application running on the same Node.

4) RIC DEPLOYMENT SCENARIOS

There is a number of different RIC deployment scenarios to host the same given ML application. For example, one operator may choose to run an application in the Near-RT RIC in a sub-1s control loop, whereas another operator may choose to run it in the Non-RT RIC where the focus is on longer trends and on reconfigurations.

5) INFERENCE HOST OPTIMIZATIONS

Host optimizations are required since the inference (usually running in the Near-RT RIC) has limited computing capability. The model selection should account for the trade-off between the efficiency and accuracy.

B. AI/ML LEARNING APPROACHES AND USE CASES

In general, there are three types of learning approaches in the AI/ML world. Next we provide some insights into how the different approaches can be leveraged in O-RAN and present different use cases such as energy efficiency, RAN slice assurance, QoS/QoE per application/user, etc.

1) SUPERVISED LEARNING

Supervised learning is an approach that uses input training data that includes labels that provide the correct result. The goal is to learn the mapping function from the input to the output from the given labeled dataset. Examples of supervised learning include: regression (linear, logistic, etc.), classification (e.g., k-nearest neighbor), decision trees, Support Vector Machines (SVM), Bayesian algorithms (Naive Bayes), ensemble algorithms, etc.

Supervised learning can be used in O-RAN such as: an rApp that provides traffic prediction and leverages regressions; clustering of RAN nodes based on performance or other parameter, which can then be used in classifications.

2) UNSUPERVISED LEARNING

Unsupervised learning is an AI/ML approach that uses unlabeled input data so that there is no known result. The task is to learn a function which can describe some hidden structure in the data. Examples of unsupervised learning include k-means clustering and principal component analysis (PCA). Unsupervised learning use cases include PCA in anomaly detection which is critical for network slice assurance.

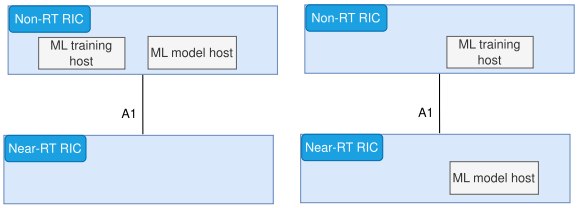


FIGURE 7. Supervised/unsupervised learning model training and host/actor locations [51].

3) REINFORCEMENT LEARNING (RL)

RL is a trial and error approach in which an agent attempts to optimize an objective by interacting with the environment. The generic RL framework is defined by a 4-tuple: 1) S: the state space of an environment; 2) A: the set of actions an agent can select from; 3) R: the reward function; and 4) T: the transition probability function.

In model-based RL algorithms, the reward and transition probabilities are based on a predictive model of the environment, and the agent uses the model to discern what would happen if a given action is taken. Model-free RL is useful in situations where deriving a model might not be possible or is not of interest. In this case, the agent does not have knowledge of the dynamics of the environment and the way an action can influence it; instead, the approach relies completely on exploration and exploitation.

RL has proven to be efficient in xApps for making decisions in sub-1s frame [52]. Researchers have also shown the applicability of RL in real-time RIC [19]. O-RAN applications that benefit from RL include network slicing, QoS resource allocation, traffic steering, etc.

4) FEDERATED LEARNING (FL)

FL is a distributed learning approach where multiple entities collaborate in the training of a model. In FL the training data is completely decentralized, and the models are trained locally, close to the data source. The local models are then aggregated in a central server. A type of FL is peer-to-peer, which uses a distributed alternative that avoids the central server.

Within O-RAN, the Non-RT RIC can serve as the central server for FL, which is then connected to multiple distributed Near-RT RIC units that serve as local FL nodes, as depicted in Figure 8. First, the Non-RT RIC distributes its ML model to its connected Near-RT RICs. Assuming that the ML training data is received through the E2 interface, the Near-RT RIC accesses the data and uses it to train the local model. The training data used by the Near-RT RIC is not shared with the Non-RT RIC, to address security and privacy considerations. Instead, the local model weights and gradients are transferred to the Non-RT RIC, which uses these to update the central ML model. Approaches for doing the update include Federated Averaging (FedAvg) or Federated Stochastic Gradient Descent (FedSGD). The exchanges between central and distributed nodes continues until the model converges. The Non-RT RIC then sends the final updated weights to the Near-RT instances [51].

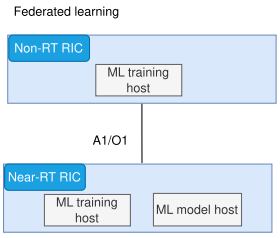


FIGURE 8. Federated learning (FL) model training and host/actor locations [51].

FL can be used in a number of O-RAN use cases. In Multi-Operator Core Networks (MOCN) multiple operators to use a single RAN; FL can provide the data privacy and security by only sharing weights with a central server that is accessed by multiple operators. Other use cases include power and computing resource allocation, and privacy and sovereignty for private networks in Industry 4.0 and V2X.

There are multiple AI/ML research challenges and opportunities in O-RAN networks. Various ML training research efforts are addressing the accuracy, generalizability, scalability, and reliability of the models. *Improving and advancing the AI modelling* and training can be tackled by investigating perception, reasoning and understanding to help the models handle complex tasks using expert knowledge. *To improve the trust in AI models*, especially when deployed in complex network systems, research needs to provide ways to ensure that the output of the ML model and subsequent actions are aligned with the goal and inspire high confidence. And finally, *scalability* should allow models to be implemented to larger and more complex problems, as well as reduce the time and complexity to implement them.

C. AI/ML LIFE CYCLE AND INTERFACE FRAMEWORKS

In order to support AI/ML applications, the O-RAN architecture (RICs and SMO) will need to host a framework such as the one shown in Figure 9 which includes data collection, preparation, training, validation, inference, and model management. We proceed to elaborate on all the components and the AI/ML lifecycle.

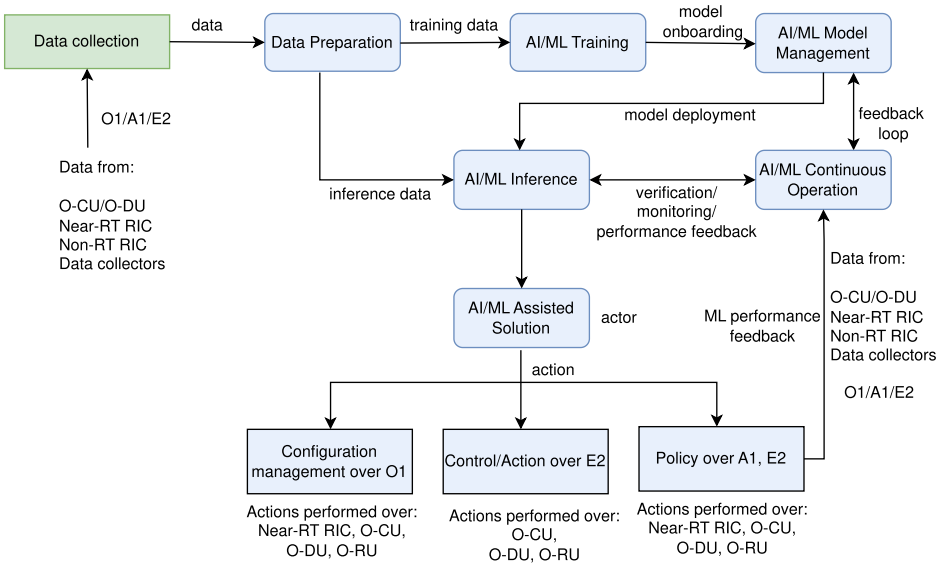


FIGURE 9. Example of an ML pipeline for O-RAN.

1) ML MODEL AND INFERENCE HOST DISCOVERY

ML model and inference host discovery is a procedure to identify the capabilities of joining nodes as ML training and/or inference hosts. These capabilities include: 1) hardware specifications of the nodes, i.e., CPU, RAM, GPU; 2) supported ML models and engines, for example, Open Network Automation Platform (ONAP) [49] components and data formats, and SDKs for different ML approaches; 3) NFVI (or CaaS) architecture [53] to run the ML models; 4) data collection and availability, and storage capabilities; and 5) inference host capabilities to support different ML models.

2) ML MODEL TRAINING

ML model training includes ML selection as well as training. This procedure resides in the SMO/Non-RT RIC, and starts with the use of the training data and other relevant metadata in the training host. Once a specific model is trained and validated, it is stored in a catalog.

3) ML MODEL SELECTION

Model selection is performed once the training and validation phase is done, and the model is ready to be deployed in the production network. Deployment viability depends on many factors, including the model accuracy and inference host capabilities. Once these factors are checked and met, deployment can be initiated through the SMO/Non-RT RIC.

4) ML MODEL DEPLOYMENT AND INFERENCE

Deployment and inference is done after an ML model is selected for a specific use case. The model is deployed, usually via a containerized image, to the inference host. The deployment, which is done through the O1 interface, should include all necessary inference host configurations for the specific model and requirements. Once the model is deployed, online data collected through E2/O1 as well as enrichment information from the Non-RT RIC over the A1 interface, are used for inference. Based on the output, various actions can be taken: configuration changes over the O1 interface; policy management over the A1 interface; control actions and policies over the E2 interface.

5) ML MODEL PERFORMANCE MONITORING

Performance monitoring may be required when feedback is needed from the inference hosts pertaining to ML model performance (accuracy, run-time, KPIs, hardware resources). This feedback may indicate to the SMO/Non-RT RIC that an update or model re-selection may be needed to improve the performance.

6) ML MODEL RETRAINING

Model retraining is performed when the feedback from the inference hosts indicates that updating and retraining will yield better performance. This usually occurs when the environment dynamics change, and the data becomes significantly different.

7) ML MODEL RE-SELECTION

ML model re-selection is performed when feedback from the inference hosts indicates that the current ML model implementation is sub-optimal and a different ML model will yield better results. Ideally, the ML catalog in the SMO/Non-RT RIC already includes other trained and validated models, and this step only require picking the next model to deploy in the inference host.

8) ML MODEL TERMINATION

Finally, ML model termination takes place when the feedback received from the inference hosts indicates that the current ML model implementation is deteriorating the network performance.

D. O-RAN DEPLOYMENT OPTIONS

We now provide an overview of the possible placements of ML components. The O-RAN Alliance has defined 4 different AI/ML pipeline deployment cases [51] according to how the components are distributed between the SMO, Non-RT RIC, and Near-RT RIC. To decide which pipeline to deploy, operators must consider the specific requirements of the rApps/xApps deployed, the resources available, data integrity and security, and so forth.

The first deployment option, depicted in Figure 10 has the Non-RT RIC as the host of the entire pipeline. Policies are pushed down to the Near-RT RIC through the A1 interface, or to the E2 Nodes through the O1 interface, depending on the control action type. This option is simple to implement, and requires low computing capabilities from the Near-RT RICs; however, it does not offer agility and flexibility for the creation of xApps.

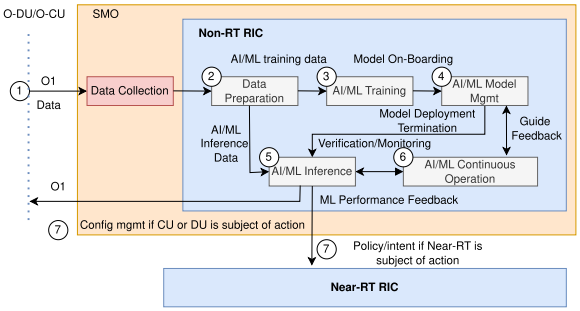


FIGURE 10. AI/ML deployment option 1: Non-RT RIC hosts the modules for continuous operation, model management, data preparation, training and inference.

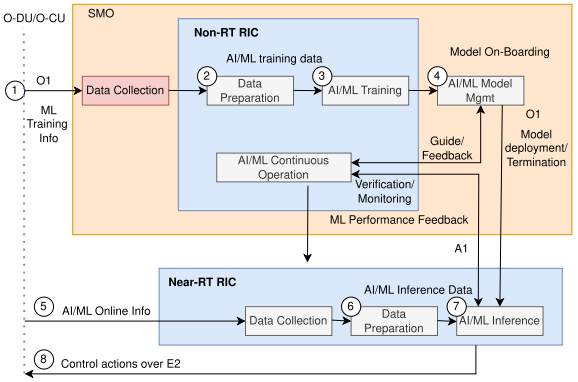


FIGURE 11. AI/ML deployment option 2: Non-RT RIC hosts the modules for continuous operation, model management, data preparation for training, and training. Near-RT RIC hosts the data collection for inference, data preparation, and inference.

Figure 11 depicts the second deployment option, which provides more flexibility and agility in terms of the ML components. Here, the Near-RT RIC hosts the inference and data collection/preparation for inference, making it more suitable for the shorter control loop requirements of the xApps. Note that there is a distinction between components such as data preparation and training running in the Non-RT RIC, and components such as the data collection and AI/ML model management that can reside in the SMO. Furthermore, there is a case to be made that the Non-RT RIC could be more decentralized and not reside within the SMO, thus making this demarcation significant for operators. The separation options of SMO and Non-RT RIC are under investigation by industry (including O-RAN WG1) and research community.

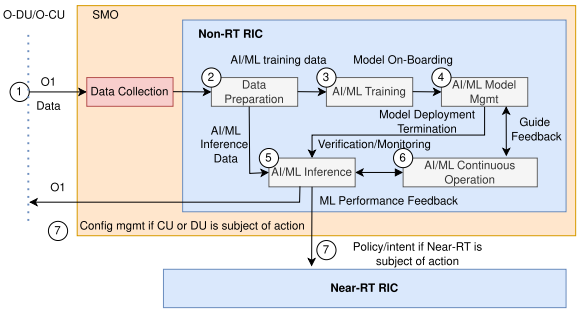


FIGURE 12. AI/ML deployment option 3: Non-RT RIC hosts the modules for continuous operation, and inference. SMO (or other entity) hosts the modules for data preparation, training, and model management.

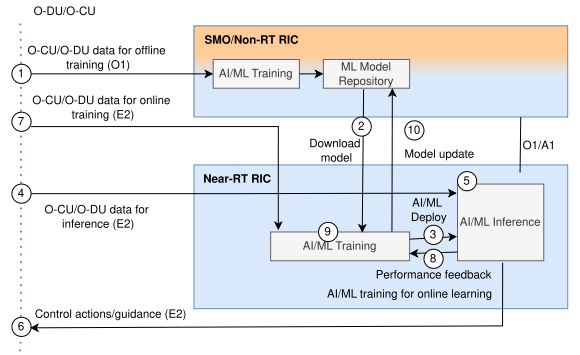


FIGURE 13. AI/ML deployment option 4: Non-RT RIC/SMO hosts the training and a repository of available models. The Near-RT RIC can perform the training (first option allowing that), and inference.

The third deployment option, shown in Figure 12, is similar to deployment option 1 in that the Near-RT RIC is excluded as the host of modules and only responsible for following the policies/intents coming through the A1 interface. Unlike option 1, in which the Non-RT RIC hosts the entire pipeline, this scenario has the SMO responsible for data collection, preparation, training, and model management, whereas the Non-RT RIC is responsible mostly for inference. The difference between option 1 and option 3 becomes greater with the separation of SMO and Non-RT RICs.

Finally, the most recently proposed deployment option, depicted in Figure 13, provides the additional flexibility to perform model training (in addition to inference) in the Near-RT RIC. The SMO and Non-RT RIC can be co-located or

not, and are responsible for the ML training and management. The placement of the pipeline components is not specified, and the choice is left to the operators to accommodate their architectural preference and specific constraints. Overall, this option offers more flexibility and agility, and possibly better performance, since the data can be collected through the Near-RT RIC and the Non-RT RIC/SMO, and training and control actions can be performed in both RICs.

VI. O-RAN RESEARCH PLATFORMS

Research platforms that leverage virtualization and softwarization facilitate faster and less costly innovation as well as open source development. Experimental research using these platforms are proving pivotal in the investigation and evaluation of the fundamental building blocks of O-RAN.

RAN (aka E2) Nodes have been *fully softwarized* for more than a decade through the efforts of open-source communities such as OpenAirInterface [54] and srsRAN [55]. These implementations are aligned with the standards specifications coming from 3GPP and provide an easy, inexpensive way to create a full-stack network using general-purpose hardware for the RAN and relatively inexpensive Software Defined Radios (SDRs) for the RU (or Radio Resource Head, RRH) [56].

Openness has allowed the wireless community to speed up innovation by offering more efficient ways to collaborate using public research platforms. These platforms offer large-scale implementation and experimentation using software frameworks, AI/ML algorithms, and approaches that are developed in academia and research labs. In particular, O-RAN research platforms require the following resources: 1) computing resources for the software stacks, such as CPU, RAM, storage, GPUs, and TPUs; 2) radio resources leveraging SDRs [56] to send/receive data; 3) RAN stacks implemented on the general-purpose hardware [54], [55]; 4) O-RAN open-source software developed by the Open-Source Community (OSC) in O-RAN [57], Open Networking Foundation SD-RAN [58], or FlexRIC by Eurecom [59], of which the OSC is the most aligned with the O-RAN standards and provides both near real-time RIC and non-rt RIC; and 5) AI/ML frameworks and approaches to the creation of the xApps/rApps for the O-RAN RICs.

Several experimental platforms offer very large-scale O-RAN experimentation. Colloseum [60] is a large network emulator consisting of 256 USRP X310 SDRs, able to capture and reproduce different wireless channel conditions. It also provides researchers with servers and GPUs, convenient for training large AI/ML models, for Near-RT and Non-RT RIC use cases. Arena [61] is an indoor research platform consisting of 64 SDRs which are controlled by software running on general-purpose hardware. The U.S. National Science Foundation (NSF) has founded the Platforms for Advanced Wireless Research (PAWR) program [62], [63]. PAWR experimental testbeds include: 1) COSMOS [64] - focusing on mmWave; 2) POWDER [65] - focusing on sub-6GHz deployments; and 3) AERPAW [66] - focusing on aerial deployments. Table 2 summarizes the description of these large-scale wireless research platforms, and the O-RAN research and testing capabilities.

The most notable European large-scale experimental evaluation and collaboration is done by 5Genesis [68], a consortium that focuses on the providing the foundational building blocks for end-to-end 5G testing (including the RAN domain and O-RAN aligned efforts) [69]. It consists of 5 testbeds: 1) The Athens Platform - focusing on edge-computing-enabled shared radio infrastructure (gNBs and small cells) for secure content delivery and low-latency applications in large public-events; 2) The Málaga Platform - focusing on automated orchestration and management of different network slices over multiple domains in the lab and in outdoor deployments; 3) The Limassol Platform - focusing on radio interfaces of different characteristics and capabilities, combining terrestrial and satellite communications; 4) The Surrey Platform - focusing on multiple radio access technologies (RATs) that can support massive Machine Type Communications (mMTC), including 5G NR and NB-IoT, combined under a flexible Radio Resource Management (RRM) and spectrum sharing platform to showcase massive IoT services; and 5) The Berlin platform - focusing on ultra dense areas covered by various network deployments, ranging from indoor nodes to nomadic outdoor clusters, coordinated via advanced backhaul technologies to showcase immersive service provisioning.

VII. END-TO-END NETWORK ORCHESTRATION ARCHITECTURE

In this section, we consider how the O-RAN architecture fits into the E2E network architecture, which also includes the transport network, the 5G core (5GC) network, and the underlying CaaS infrastructure. We will be especially focused on E2E network slice orchestration since network slices will predominate in providing the communication for future applications such as Industry 4.0 and V2X [1], [70], [71]. The E2E network slice orchestration framework includes components and interfaces that are defined through different standards development organizations (SDOs), such as the 3rd Generation Partnership Project (3GPP), European Telecommunications Standards Institute (ETSI), Internet Engineering Task Force (IETF), the O-RAN Alliance. We will review the components and provide a workflow summary of how slice instantiation involves the control, management and orchestration layers.

The E2E network slice orchestration framework consists of various domain-specific components and requirements, as well as components that provide the E2E view of the slice and orchestration in terms of performance and user experience. Different components have different scope and features and responsibilities, so in order to achieve E2E targets, the components need to communicate through standardized interfaces to exchange data, collect data, push control actions, and so on. There is ongoing work in SDOs

TABLE 2. Most notable large-scale O-RAN research platforms.

Research project/ Research platform	Platform description and research area	HTTP location
Colloseum	Large network emulator (256 SDRs), ability to tackle wide variety of research areas, including the creation and implementation of O-RAN RICs and applications. Servers and GPUs for training large AI/ML models for Near-RT and Non-RT RIC use cases.	https://www.northeastern.edu/colosseum/
Arena	Indoor platform consisting of 64 SDRs and computing equipment. Research areas include O-RAN orchestration [67], security, spectrum sensing, energy efficiency, etc.	https://ece.northeastern.edu/wineslab/arena.php
COSMOS	COSMOS consists of 9 Large Sites, 40 Medium sites, Fiber optic connection from most sites, 200 Small nodes. It is also an O-RAN Open Testing and Integration Center (OTIC), supporting the evaluation of O-RAN use cases.	https://www.cosmos-lab.org/
POWDER	Powder provides SDRs, compute, networking, and software stacks to support various research areas. Powder is a member of the O-RAN Alliance, providing reference RIC in connection with open source mobility stack. In addition, Powder adds RAN slicing functionality to the reference RIC. Powder also serves as an O-RAN Open Testing and Integration Center (OTIC)	https://powderwireless.net/oran
AERPAW	AERPAW is the first research platform to study the convergence of 5G technology and UAVs. It includes wireless equipment such as SDRs, UAVs, testing simulators, fixed and portable nodes, computing and networking env, etc. AERPAW also serves as an O-RAN Open Testing and Integration Center (OTIC).	https://aerpaw.org/otic/

to bring the modularity and interoperability required to efficiently implement an E2E network slice orchestration framework.

We now discuss the components, their roles and responsibilities, as well as their role in the E2E view. Figure 14 depicts the overall network architecture, consisting of the three distinct domains - access, transport, and 5GC network. Each domain has a set of domain-specific controllers that are responsible for their control and optimization.

A. SLICING COMPONENTS

These components have been defined by 3GPP and include three layers: the Communication Service Management Function (CSMF), Network Slice Management Function (NSMF), and Network Slice Subnet Management Function (NSSMF). In terms of responsibilities, from a top-down perspective, the CSMF receives the communication service requirements from the OSS/BSS systems, and maps these to specific network slice requirements. The NSMF connects to the CSMF, through the southbound interface, and is responsible for the management and orchestration aspects of network slice instances (NSIs), such as the breakdown of network slice requirements into network slice subnet requirements, and the distribution of those requirements to the NSSMF through the southbound interface. Finally, the NSSMF is responsible for the management and orchestration of the network slice subnet instances (NSSIs), which are received from the NSMF through the northbound interface. The NSSMF then communicates with the domain-specific controllers through the southbound interface.

Consider a use case such as RAN slice SLA assurance. The RAN Network Slice Subnet Management Function (R-NSSMF) is responsible for the performance and lifecycle of the subnet, based on the requirements coming from the network slice management function (NSMF) which is hosted in the service management and orchestration framework (SMO). Thus, the R-NSSMF will translate the requirements (namely, latency, throughput, reliability) coming from the northbound interface (NBI) into actionable items to optimize the resources of the RAN (e.g., prioritize users or slices, allocate more resource blocks (RBs) to specific users or slices, etc.), while focusing on the specific slice requirement. The same approach is applied to the Transport Network Slice Subnet Management Function (T-NSSMF) and Core Network Slice Subnet Management Function (C-NSSMF), which control the resources of the transport and core network, respectively. Finally, at the E2E level the SMO hosts the Network Slice Management Function (NSMF) the and Communication Slice Management Function (CSMF) that are responsible for the end-to-end requirements of the slices [23], [43], [45]. The placement of the NSSMFs is decided subject to the specific requirements and trade-offs of operators.

B. DOMAIN DATA ANALYTICS FUNCTIONS (DAFS) AND CONTROLLERS

Domain-level orchestration is key to meet the communication requirements of new slice types. We will provide insights into the capabilities and requirements of data analytics functions (DAFs) and controllers in each domain. We will then provide the E2E slice orchestration view.

1) NETWORK DATA ANALYTICS FUNCTION (NWDAF)

NWDAF is a 3GPP network function [72] that aims to provide real-time closed-loop operational intelligence in the 5GC. Unifying the data from multiple sources provides a distinctive opportunity to provide correlated insights from 5GC service-based interfaces, logs from the user and

control plane, data from the UE, and application-level data. The analytics opportunities include network performance, network function load, quality of service (QoS), slice load level, and anomaly detection. NDWAF may play a central role in O-RAN use cases such as RAN slice SLA assurance, since the application-level QoE metrics are external (EI) information that will likely come from the NWDAF or a similar source.

An NWDAF fully compliant to the 3GPP specifications [72] is still under development. However, cloud-native open-source 5GC solutions leverage tools such as ElasticSearch [73] and Prometheus [74] to provide monitoring of performance metrics, interface logs and configuration metrics that could be used in 5GC analytics and closed-loop automation.

2) TN DAF AND CONTROLLERS

TN DAF and controllers provide seamless connectivity between the RAN and 5GC components. For data collection, the objective is to provide monitoring, visibility, and observability of the underlying infrastructure. The general approach to develop DAF capabilities is to leverage advanced techniques and ML-based solutions to optimize network performance and maximize the resource utilization, while achieving the real-time QoS/QoE requirements of different slices.

Transport network controllers such as Jalapeno [75] provide tooling to gain network visibility by collecting telemetry and topology data, and to create cloud-native applications to control the network. Other controller solutions include ONOS [76], and OpenDaylight [77].

3) RAN DAF

RAN DAF is a functionality to create data warehouses in the RICs or SMO, with the goal to unify data types coming from different interfaces (E2, O1, O2, A1), and to process the data for analytics use cases. As described in Sections III and IV, the RICs provide intelligence via ML-based management and orchestration of resources [78]. The RAN DAF functionalities are becoming native to the RICs and to the applications built on top of them to provide optimized performance [51]. RICs can operate in two control loops, near-rt which is 10ms to 1s and non-rt which is greater than 1s.

4) CLOUD DAF

For analytics pertaining to cloud performance, namely Cloud DAF, the most popular CaaS solutions such as Kubernetes [53] and OpenShift [79] offer tools to monitor and collect infrastructure-level data that can be analyzed and controlled to provide infrastructure assurance.

C. SERVICE MANAGEMENT AND ORCHESTRATION

The SMO or E2E Orchestrator is the framework responsible for the oversight of the E2E slices. Multiple components in the SMO framework provide the management and orchestration capabilities to network domains and E2E slices. In general, the SMO hosts the 3GPP slicing components, CSMF and NSMF. The SMO maps the communication requirements that are provided by the OSS/BSS systems to slice-specific requirements. Furthermore, the NSMF (usually hosted in the SMO) takes those slice requirements and provides the per domain breakdown. The domain NSSMF can either be hosted by the SMO as well, or be external. The NSSMF would then take those domain-level slice SLAs and use the controllers to take actions as necessary.

Typically, the SMO hosts the Non-RT RIC. As discussed in previous sections, V-D, the SMO and Non-RT RIC are normally tightly coupled. However, there are instances where that might not be the case. For example, an operator may prefer a highly modular architecture in which the RIC is provided by another vendor. The SMO and RIC frameworks need to be implemented in a way that is aligned with standards and provides that modularity and interoperability.

There are two well-developed open-source solutions, that are considered as the SMO for the RIC development in the O-RAN alliance. Open Network Automation Platform (ONAP) [80] is a mature project developed by the Linux Foundation. It's highly complex and modular, running in operator networks as the SMO. Currently the O-RAN Alliance RIC (OSC) is integrated with ONAP as the SMO.

The second open-source SMO project is Open Source MANO (OSM) [50], hosted by ETSI and aligned with the ETSI NFV specifications [81], [82], [83]. OSM is less computationally demanding in comparison to ONAP; however, it was not as adopted by the industry. OSM is the next SMO under consideration for integration with the OSC RICs.

Finally, Figure 14 provides an architecture for end-to-end orchestration of network slices, along the lines of the recent project called Nephio [84]. Nephio is placed between the SMO (E2E Orchestrator) and the domain-level controllers, and it is tasked with the lifecycle management of network functions implemented on top of a Kubernetes-based CaaS. Nephio communicates with the SMO to exchange cloud platform data (through the O2 interface) and to push policies from the SMO to the different domains.

VIII. O-RAN USE CASES

In this Section, we will present a subset of use cases that have been defined by the O-RAN Alliance, especially focusing on SLA assurance, resource allocation, traffic steering, and energy efficiency [20], [85]. The common denominator of these use cases is the focus to achieve slice assurance in an energy and resource efficient manner.

First, we present notable O-RAN use cases in Table 3 which gives a brief overview, including the input data, ML output actions, and placement of the ML training and inference on the RICs. Suppose we take the example from Table 3, for the radio resource allocation to unmanned aerial vehicles (UAVs). The goal of this use case is to perform

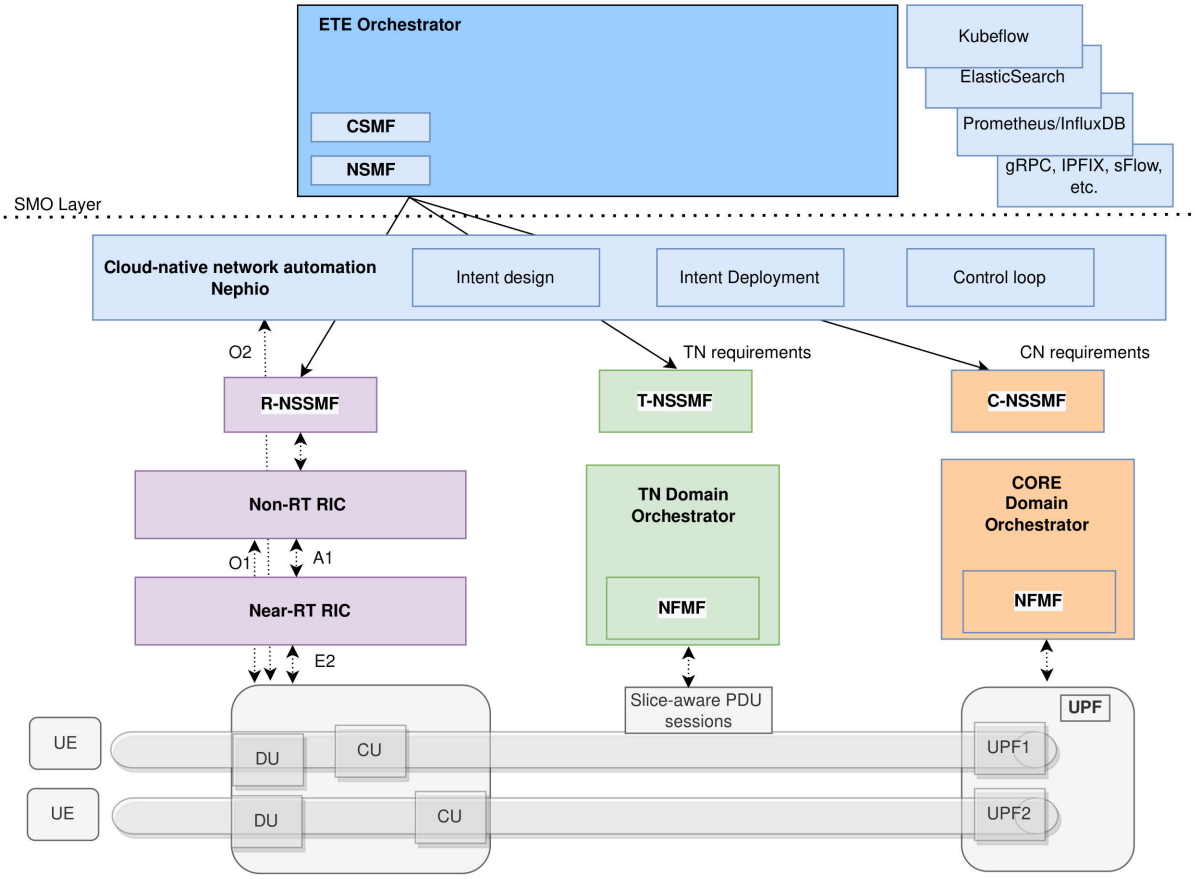


FIGURE 14. Architecture for end-to-end network slice orchestration.

radio allocation to UAVs to overcome interference and poor KPIs, at the MAC layer where the changes are invoked. To configure the RAN nodes in a data-driven manner, we need to collect and analyze data such as UE measurement reports, radio channel information, flight path, 3D area information, etc. Due to latency constraints, the Non-RT and Near-RT RIC need to be invoked as training and inference host, respectively.

Next we take a deep dive into several use cases.

A. RAN SLICE SLA ASSURANCE

O-RAN [22] defines this use case for a 5G system to support the user demands of slices and their requirements (i.e., data rate, latency, reliability, availability, user density, etc.) offered through the same underlying infrastructure. The requirements are specified in the Service Level Agreement (SLA) between the operator and the customer, and it is in the interest of both parties to ensure that the slice SLAs are respected and that violations are minimized. O-RAN, through the open interfaces and ML platforms, offers the opportunity to create such protective mechanisms and effectively implement the lifecycle management of the network slices.

The RAN slice SLA assurance use case involves the Non-RT RIC, Near-RT RIC, RAN (E2 nodes), and SMO interactions. The Non-RT RIC provides monitoring and long-term trends and pattern recognition for the RAN slice subnet performance. It also provides the MLOps (from data processing to model training and validation) with the output being reconfigurations through the SMO (via the O1 interface) or through the Near-RT RIC (via the A1 interface). The Non-RT RIC can also supply the trained model to the Near-RT RIC, thus providing ML inference residing on the Near-RT RIC side and operating in the sub-1s time scale.

As with the Non-RT RIC, the Near-RT RIC also provides monitoring of specific RAN KPMs; however, at a different time scale. As mentioned above, the Near-RT RIC is envisioned to host the ML inference component to deploy and execute models that have been pre-trained in the Non-RT RIC. A third important functionality of the Near-RT RIC is the ability to interpret and execute A1 policies coming from the Non-RT RIC (northbound communication), and subsequently perform the optimized RAN actions (through the E2 interface, southbound communication) to achieve the requirements that are based on the O1 configuration, the A1 policies, and the E2 reports.

For this use case to be attainable: the RAN (E2 nodes) need to be able to support all the interfaces and functionalities, i.e., supply slice specific performance measurements through the E2 and O1 interfaces; And support the required actions, such as slice-aware resource allocation, prioritization, and so on.

TABLE 3. O-RAN use cases employing AI/ML as defined by the Alliance [21], [86].

Use case	Layer	ML outcome	Training data	Action	Training host	Inference host
Massive MIMO beamforming optimization	PHY	Predict the optimal configuration of mMIMO params for each cell. Operator defines global optimization objective	Calculate KPIs from: traffic load, coverage, QoS, mMIMO	Configure the optimized beam params via the O1 interface	Non-RT RIC	Non-RT RIC
Quality of Experience (QoE) optimization	MAC	Predict the traffic types/ application, and map QoS to QoE	Network-level info: UE data, L2 measurements, RAN protocol, cell-level information, QoS measurements (mapped to QoE), user traffic	E2 control or A1 policies to change RAN parameters based on QoE requirements	Non-RT RIC	Near-RT RIC
Dynamic spectrum sharing	MAC	Predict the short-term traffic demand based on near-RT metrics from the RAN	Location, DSS modality, cell configuration, PHY PRBs, number of active UEs, traffic demand, QoS classes, and UE capability information.	E2 control/policy of resource configuration and scheduling coordination.	Non-RT RIC	Near-RT RIC
Network slice subnet instance (NSSI) resource allocation optimization	MAC	Predict of the traffic demand patterns of NSSI at different times and locations.	Measurement counters per slicing subnet instance. These include DL/UL PRBs used for data traffic, avg DL/UL throughput, and the # of PDU sessions successfully set up and failed to set up.	Reconfigure the NSSI attributes via the O1 interface and update the Cloud resources via the O2 interface.	Non-RT RIC	Near-RT RIC/ Non-RT RIC
Context-based dynamic handover (HO) management for vehicular users	MAC	Customize HO sequences for a UE according to intents, policies, and configurations taking into consideration the high speeds and the heterogeneous environment of vehicular UEs	Historic data from a vehicular UE over the V1 interface, which include UE location, velocity, and trajectory, measurement reports for serving and neighboring cells, and connection and mobility statistics.	RAN deploys the configuration received from the near-RT RIC over the E2 interface.	Non-RT RIC	Near-RT RIC
Radio resource allocation to unmanned aerial vehicles (UAVs)	MAC	Perform the radio resource allocation for UAVs to tackle challenges such as RF interference, poor KPIs caused by base station antenna patters (side lobes towards sky).	Aerial vehicle related measurement metrics from SMO and based on UE measurement reports, radio channel information, mobility related metrics, flight path, climate, forbidden/limited 3D area information, etc.	The near-RT RIC delivers the radio resource allocation configurations to the O-CU/O-DU over the E2 interface.	Non-RT RIC	Near-RT RIC
Traffic steering	MAC	Traffic steering among various RATs, such as LTE, NR, and Wi-Fi, according to different traffic management objectives.	Measurement reports, cell load statistics cell quality thresholds, channel quality indicator reports and measurement gaps on per-UE or per-frequency basis.	The traffic management policy is transferred to the near-RT RIC via the A1 interface.	Non-RT RIC	Near-RT RIC

B. NSSI RESOURCE ALLOCATION OPTIMIZATION

5G brings a palette of services and applications that have very distinct characteristics and exhibit different spatio-temporal demands and patterns. For example, organized events tend to produce large amounts of traffic in specific locations; IoT sensors tend to produce steady and periodic traffic; and URLLC applications will likely solicit bandwidth, latency, reliability, availability.

The NSSI Resource Allocation Optimization use case provides for efficient resource allocation to a Network Slice

Subnet Instance (NSSI) thus optimizing for the service requirements. This is achieved through the exploitation of resource quota policies associated with the RAN nodes included in the NSSIs, which maximize resource utilization based on historical data and ML models that learn the behavior and predict the resource demands. This process involves three steps: monitoring, analysis, and execution.

The first step, monitoring, includes the collection of data via the O1 interface. These data includes performance measurements, such as:

- Downlink Physical Resource Blocks (PRBs) used for data traffic;
- Uplink PRBs used for data traffic; Average DL and UL throughput in a single gNB;
- Distribution of DL and UL throughput in gNB;
- Number of PDU sessions, both requested and setup;
- Number of Data Radio Bearers successfully setup.

The second step, analysis and decision-making, consists of two distinct components. The first component is ML models to analyze the performance measurements and forecast future spatio-temporal traffic demands for each NSSI. The second component is the choice of output actions for the RAN NFs (E2 nodes) in the NSSI, for example, spectrum allocation, VNF/CNF resources, and slice attributes.

The third step is execution. First, the NSSI attributes are reconfigured via the OAM functions in the SMO, and using the O1 interface to configure the RAN NFs. Next the cloud resources are updated through the O2 interface. The SMO employs an action for NSSI resource optimization consisting of provisioning of the NSSI resource quota policy, which serves as a constraint (upper limit boundary) for the NSSI resource allocation optimization.

The Non-RT RIC provides the following functionalities for this use case: Collection of PMs relevant for NSSI resource utilization via the O1 interface; Training of ML models based on historical data, with the goal to forecast spatio-temporal traffic demands of NSSI; Decision-making (during the inference mode) in terms of resource allocation (add/reduce) for different O-RAN nodes at different times.

In this use case, the E2 Nodes (including RU, DU, CU-UP, CU-CP) provide two functionalities. The first is performance measurement collection with a required granularity through the O1 interface. And the second is configuration and updates of NSSI resource allocation through the O1 interface.

C. QOS RESOURCE ALLOCATION

The QoS Resource Allocation use case is similar to the previous use case, and is implemented as an xApp [87]. For dynamic allocation of radio resources, the xApp controls the quota of PRBs allocated to different slices to meet their SLA requirements, while taking into consideration the user demands. The xApp targets to improve utilization of the PRB resources among different slices.

This xApp requires a number of parameters to be collected and used to optimize the resource allocation. Table 4 provides an overview of the main and optional parameters, as well as their description. It is noticeable that baseline information such as DL and UL throughput, packet loss, PDU session statistics, as well as slice-level information is necessary to provide the right level of visibility. The output actions of this use case are reallocation of resources per slice, and eventually, per user.

D. TRAFFIC STEERING

Traffic steering is one of the most important use cases for network slicing because it is expected to address the operator's struggle to efficiently use the resources in macro and small cells. For instance, to achieve eMBB throughput for users that are streaming content, it is always preferable to connect to a small cell rather than use the limited bandwidth in a macro cell. This is not simple to achieve, when there is no visibility of the available network resources, and when data-driven solutions are not possible. In addition, there are certain scenarios where avoiding frequent handover improves user experience (e.g., voice calls).

To enable intelligent and flexible gNB association, traffic steering needs to provide mechanisms to optimize the association based on targets defined by operators, for example, service-based association or energy saving. The traffic steering xApp (operating in control loops below 1s) should control the cell preferences and handovers that are related to individual users, QoS flows, or network slices.

Generally speaking, the traffic steering use case expects a set of parameters to be collected and used for training and inference. The baseline and optional parameters that are used as input to the decision-making for this use case are included in Table 5. Similar to the previous use case, the input parameters include statistics about the QoS metrics, as well as slice-level information. Additional parameters include mobility management as well, as the traffic steering can be performed inter- and intra-gNB and frequency bands.

The Traffic Steering xApp must provide closed-loop control and automation of RAN parameters, without human intervention. The first parameter that is under control of the application is the *UE ID*, which is used to specify which UE is subject to handover. Since the objective of the xApp is to steer different users to different cells based on the type of service/application that the user requires, the second parameter under control is *Primary cell ID* which CGI of the primary cell that serves the UE selected for handover. *Target primary cell ID* is another parameter that the xApp should control since it represents the cell that is target cell in the handover procedure. To provide seamless service, *list of PDU sessions for handover* is a parameter which includes PDU session identifiers subject to handover, also under control of the xApp. And the last envisioned parameter to control is *list of DRBs for handover*, which includes the data radio bearer identifiers subject to handover.

E. ENERGY EFFICIENCY

Traffic demand varies in a daily, weekly, and monthly manner due to the seasonal pattern of customer mobility

TABLE 4. Input parameters and description for the QoS resource allocation use case.

Input parameters	Parameter description
Distribution of DL and UL throughput in a gNB	Information about the UE at multiple granularities, i.e., within a cell, slice, and flow
Radio Resource Util	Includes measurements such as: DL/UL total available PRBs in a certain cell; mean DL/UL PRB used for data traffic over time, per cell, slice, or flow; peak DL/UL PRBs used for data traffic over time per cell, slice, or flow.
Number of active UEs	Can be obtained by: max number of active UEs in DL/UL per cell, slice, or flow; and number of active UEs in DL/UL per cell, slice, or flow.
PDU session management	Includes the following components: number of PDU sessions requested to setup within the gNB or per slice; number of PDU sessions successfully setup within the gNB or per slice; and number of PDU sessions failed to setup within the gNB or per slice
Cell Global Identity (CGI)	A combination of: Public Land Mobile Network Identity (PLMN ID) and E-UTRAN Cell Identity (ECI) or New Radio Cell Identity (NCI)
Single - Network Slice Selection Assistance Information (S-NSSAI)	Used to identify network slices within a PLMN, a combination of slice type (SST) and slice differentiator (SD)
(Optional) Distribution of DL/UL packet drop/loss rate	Used to provide the distribution of UL and DL packet loss in a cell, slice, QoS flow, etc.
(Optional) Distribution of DL/UL delay between NG-RAN and UE	Used to provide the distribution of the E2E delay between RAN and UE, with granularity of cell, slice or QoS flow.
(Optional) 5G QoS Identifier (5QI)	Provides the QoS characteristics that define the requirements for a QoS flow, e.g., packet error rate, priority, etc.

TABLE 5. Input parameters and description for the traffic steering use case.

Input parameters	Parameter description
Distribution of Synchronization Signal Reference Signal Received Power (SS-RSRP)	Provides information about the SS-RSRP received by the gNB from the UE, from both serving and neighboring cells
UE ID	Used to uniquely identify the user subjected to handover
Cell Global Identity (CGI)	Combination of Public Land Mobile Network Identity (PLMN ID), and E-UTRAN Cell Identity (ECI) or New Radio Cell Identity (NCI)
S-NSSAI	Used to identify network slices within the PLMN
5G QoS identifier (5QI)	Used due to the significance of the QoS characteristics that define the requirements for a QoS flow, e.g., packet error rate, priority, etc.
Mean number of RRC connections	Provides the mean number of active UEs (mean number of users in RRC connected mode for each NR cell during some period)
Max number of RRC connections	Parameter that provides the information about the max number of active UEs
(Optional) Distribution of DL/UL UE throughput in gNB	Provides the xApp with information about the UE at multiple granularities, such as within a cell, slice, or QoS flow
(Optional) Radio resource utilization	Can be ascertained using the: DL/UL total available PRBs in a certain cell; mean DL/UL PRB used for data traffic over time, and per cell, slice, or flow; peak DL/IL PRBs used for data traffic over time, and per cell, slice, or flow.
(Optional) Mobility management	Includes the following measurements: inter-gNB handovers - # of requested handovers, and # of successful handovers; intra-gNB handovers - # of requested handovers and # of successful handovers; intra/Inter-frequency handover - between freq bands within the same (intra), or the neighboring (inter) cells.

and behavior. For example, business hours during workdays place strain in the downtown infrastructure where most offices are located. However, on evenings and weekends, the demand is more distributed in the suburban areas. The spatio-temporal variation in demand creates an opportunity to switch off entire cells or turn off some of operating frequencies to provide energy saving. The energy efficiency use case addresses the extensive energy requirements in O-RAN deployments through the dynamic switching on-and-off of cells or frequencies according to demand.

An rApp for energy efficiency optimization requires a long-time-scale (minutes, hours, and higher) forecasting. It runs on top of the Non-RT RIC to provide a data-driven on/off switching of cells. The information flow in this use case is as follows. First, the E2 Nodes are configured by the Service Management and Orchestration (SMO) to report the data necessary for energy-saving algorithms via the O1 Interface to the Collection and Control unit. Assuming that the Non-RT RIC and SMO are tightly coupled the Non-RT RIC retrieves the collected data through internal SMO communication.

The O-RUs are involved in this use case. The E2 Nodes need to configure them to report data through the Open RAN Fronthaul Management Plane (Open FH M-Plane) interface. The energy efficiency rApp takes into account two main components: 1) Quality of Service (QoS) of the users; and 2) energy consumption of the system. To fulfill the QoS requirements for users while trying to minimize for the energy consumption, the following performance measurements need to be collected: per cell (per carrier) power consumption, user throughput (mostly downlink at this moment), users traffic type (i.e., 5G QoS Identifiers), per slice radio resource utilization, etc. Once the data reaches the Non-RT RIC, a preliminary analysis is conducted which provides long-timescale statistics of cell load, user throughput, and related power consumption. These statistics are used to train the ML model, and after the training and validation, the model is deployed (i.e., ML inference stage).

The output of the ML inference is the decision-making of switching specific cells on/off. After the decision is made, the Non-RT RIC sends the execution request to the Collection and Control Unit via internal SMO communication. The next step is the configuration of the E2 Nodes through the O1 interface, which is done by the Collection and Control Unit. The E2 Nodes proceed with the configuration down to the related O-RU. To provide the best accuracy and performance of the deployed ML model, the best practice is to have a feedback loop from the ML inference to the ML management layer. In this way, if the conditions in the network change and there is detected degradation of the performance, the model can be retrained to take into consideration the new conditions, and then redeployed.

Prior to O-RAN, the research community had proposed approaches for energy conservation, and this work [88], [89] is relevant in the context of O-RAN and can be implemented in rApps. The authors in [88] have proposed extending the network architecture with a Radio Environment Map (REM) database, to map user locations on the energy efficiency related to a particular set of active cells. The REM is used as data storage for a Reinforcement Learning (RL) agent that interacts with the environment to learn which set of active cells will maximize the energy efficiency for a given group of users.

Additional rApps can be deployed as the research community creates additional intelligent applications that enhance the performance of O-RAN deployments. For example, works like [88] should find their way to the O-RAN architecture, and provide an rApp that can leverage RL approaches to optimize the energy efficiency.

IX. RESEARCH CHALLENGES

Research efforts are very active in the O-RAN community, focusing on: 1) architecture and capabilities of the RIC; 2) xApp/rApp development toolkits, from data accessibility to ML approaches; and 3) openness of interfaces, especially regarding E2, A1, R1, O1. In this section we discuss research challenges that are directly related to development of efficient RICs, and to capabilities related to orchestration of network slices. Most of these challenges are applicable to all O-RAN use cases.

A. CONFLICT MITIGATION

The simultaneous deployment of multiple xApps and rApps for a variety of applications makes it likely that conflicting application objectives will need to be resolved by the Near-RT and Non-RT RICs. Conflicts result when a reconfiguration triggered by one application results in performance degradation of another application. For example, an operator running an energy efficiency application may attempt to minimize energy consumption while satisfying the QoS requirements for its users. On the other hand, a QoS resource allocation or QoE management application may have the objective to optimize user experience. Both applications might attempt to change the modulation and coding schemes (MCS), with the first one seeking a lower and the latter seeking a higher MCS to achieve their respective objectives. Thus a conflict mitigation module is introduced to resolve conflicts that emerge among different xApps and rApps in the Near-RT and Non-RT RIC.

The scope of the conflict may be tied to a user, a bearer, or a cell, and it can be related to a configuration or action performed by the RIC. Generally speaking, there are three different classes of conflicts: direct, indirect, and implicit conflicts.

Direct conflicts can be directly detected by the conflict mitigation component in the RIC. For example, multiple xApps may apply different configurations for the same parameters in the same control target, or they may request more resources than are available. These conflicts can be solved by the conflict mitigation component that decides which xApp has precedence and then limiting the scope of a control action.

The *indirect and implicit* conflicts cannot be observed directly, that is, it is difficult to find the relationship and causality among different xApps. An example for these types of conflicts are configurations that optimize the performance of certain classes of users, but degrade others in non-obvious ways. These conflicts may be detected and mitigated through verification and monitoring of the performance of the system after the application of control policies.

While conflict mitigation is a key component of the RICs, it is still in the R&D stage at this time with works such as [90], [91], and [92], and is still not included in the OSC implementation of the RIC. In addition, once Non-RT RIC and Near-RT RIC conflict mitigation modules are implemented, there will be a need to synchronize and possibly create a hierarchical conflict mitigation system to deal with the inter-RIC interactions. Furthermore, the emergence of a real-time RIC will add an additional layer of complexity that needs to be tackled by a hierarchical approach.

B. DATA SHARING, MANAGEMENT AND EXPOSURE

The O-RAN Alliance use cases [93] include scenarios in which the xApps and rApps need to collaborate and share

data to optimize a certain objective. An example is RAN slice SLA assurance, which combines both types of applications operating in different control loops and consuming data coming from the E2, O1, and O2 interfaces. Currently O-RAN WGs are looking into ways of sharing data between the Near-RT RIC and Non-RT RIC through the A1 interface. An example driving this work is an rApp that needs E2 data that O1 and O2 would not be able to collect. Thus, creating the appropriate mechanisms for data management and exposure for the RICs is critical.

Another example requiring xApp and rApp collaboration is the QoE optimization scenario from O-RAN [93] which assumes inputs from external systems and monitoring of application performance. Thus, multi-dimensional data types should be combined to identify the best RAN configuration for the optimization of the QoE per user. As in the previous example, the O-RAN interfaces become very important in handling heterogeneous data combination and fusion to bring additional information to optimize QoE. This was not possible in traditional RAN deployments.

C. REAL-TIME RIC DEVELOPMENT

Some of the lower-layer control functions need much faster control loops than the near-RT RIC is able to deliver. A real-time RIC would be implemented on the CU or DU side, with the goal to provide control actions in sub-1ms time scale. Researchers have already provided some guidance on this. For example, D'Oro et al. have introduced dApps [18] that receive real-time data from the RAN and EI from the near-real-time RIC and execute inference and control of lower-layer functionalities. These dApps can address use cases with stricter timing requirements than those considered by current RICs, for example in beam management and user scheduling.

Another real-time RIC is implemented in [19] where EdgeRIC is collocated with but separate from the E2 Nodes. EdgeRIC is shown to significantly outperform a cloud-based near-RT RIC in terms of throughput, and decision-making can be done in much more stringent manner, at a transmission time interval scale.

Further Investigation is required to determine the trade-offs in massive scale deployment of real-time RICs in terms of use case necessity vs. practical limitations (computing resources and conflict mitigation).

The O-RAN community needs to provide more guidance on the real-time RIC. For example, the optimal placement of the RIC and its integration in the O-RAN architecture could be done by leveraging and extending interfaces and components that have already been standardized. However, other considerations may suggest otherwise. Thus a careful investigation should consider proposals for architectural changes to accommodate real-time RICs.

A closer look at the use cases for the real-time RIC is needed. There are 5G slice types such as URLLC that impose very stringent communication and reliability requirements that can only be achieved with extremely tight control and optimization. Furthermore, ongoing 6G use case definition will introduce extremely stringent requirements, and the standards gap analysis will unavoidably lead to the creation of a real-time RIC WG. Other considerations on hardware acceleration, virtualization technology, management and orchestration functionalities, and so on will need to be tackled, either as part of the existing WGs or as part of a new WGs within O-RAN.

D. MLOPS AND USE CASE APPROACHES

There is ongoing standardization work within the O-RAN Alliance on the ML placement options described in Section V. Multiple WGs within O-RAN are impacted by this work item, most notably WGs focusing on the RICs and SMO. Discussion points include the placement of ML pipeline components across the RICs and SMO, data sharing and exposure between components as training and inference can be done in different components, interface capabilities for data collection (namely E2 and O1), etc.

Another aspect of MLOps is the overall architecture and the integration of different, mostly open-source ML frameworks and pipelines, such as Kubeflow [94], Acumos [95], as well as monitoring systems such as Prometheus [74] and ElasticSearch [73].

Finally, as the foundation for the placement of ML pipeline components and overall architecture of the MLops solution is laid, the R&D efforts will focus on AI/ML model development, training, validation, inference, and testing, as well as the lifecycle management. Research efforts have already provided insights into the use of advanced ML approaches such as: i) reinforcement learning, especially for physical layer control; ii) federated learning, especially for collaborative learning in environments in which data security and privacy are critical; iii) deep learning, especially for forecasting and learning of long time-series. More novel machine learning approaches need to be implemented and tested to provide learning and guidelines to the O-RAN community.

In Table 6, we summarize the most notable research work that addresses the aforementioned challenges.

E. RIC AND XAPP/RAPP COMPATIBILITY

Section IV presented the non-RT RIC and highlighted the importance of the R1 interface in the mediation between the rApps and the RIC. Section III presented the near-RT RIC and highlighted the importance of the A1 interface between the two RICs, as well as the E2 interface between the near-RT RIC and the E2 Nodes. To provide plug-and-play features and allow for different xApp/rApp vendors to easily integrate with different RIC vendors, it becomes critical for vendor implementations to strictly follow the standards from the O-RAN WGs. This challenge is tackled by focus groups in the O-RAN Alliance, as shown in the next Section.

X. O-RAN ALLIANCE EFFORTS

So far, we discussed the O-RAN architecture, use cases, and research challenges. In this section we review the structure

TABLE 6. Notable research work addressing the defined challenges.

Reference	Research area	Description	Implementation
[92]	Conflict mitigation	Definition of conflict mitigation framework (CMF), built into O-RAN Near-RT RIC.	Simulation environment with 19 base stations, and O-RAN network including both RICs.
[90]	Conflict mitigation	Universal method for detecting and resolving indirect conflicts between xApps by tracking changes with Parameter Groups (PGs) that influence the same area of network operation in relation to specific control targets.	Simulation environment of macro and micro base stations, and a Near-RT RIC with deployed two xApps: Mobility Load Balancing (MLB) and Mobility Robustness Optimization (MRO).
[96]	Data sharing, mgmt, and exposure	SCOPE is an open and softwarized prototyping platform made of: 1) portable open-container for instatiating network elements (RAN and UEs), 2) emulation module for real-world channel and traffic conditions; and 3) a data collection module for AI/ML-based applications.	SCOPE implements APIs at run time to be able to implement closed-loop optimization and data-driven (e.g., throughput, MCS, slice PRBs, etc.) control loops. SCOPE is implemented on top of large-scale platforms such as Colloseum.
[60]	Data sharing, mgmt, and exposure; MLOps and use case approaches	ColO-RAN is a large-scale O-RAN testing framework, that builds on top of the capabilities of Colloseum and enables ML research at scale using O-RAN components.	Three xApps based on Deep Reinforcement Learning (DRL) are developed, deployed, and tested. To demonstrate application portability, the implementation is done on top of Colloseum, and Arena.
[63]	Data sharing, mgmt, and exposure; MLOps and use case approaches	OpenRAN Gym is an open and O-RAN-compliant experimental toolbox for data collection, design, prototyping and testing of end-to-end data-driven solutions that control the RAN in different timescales (i.e., xApps and rApps).	OpenRAN Gym is implemented on top of projects such as SCOPE and ColO-RAN. The evaluation of xApps built with the OpenRAN Gym toolbox is implemented on top of large-scale research platforms, COSMOS and POWDER.
[97]	Data sharing, mgmt, and exposure; MLOps and use case approaches	This work creates xApps that take into consideration the QoE metrics that are most significant for applications such as cloud gaming, AR, and video-enabled voice calling.	Implemented in a real-world testbed using HexRIC [98] and HexRAN [99] as the foundational blocks.
[100]	Data sharing, mgmt, and exposure; MLOps and use case approaches	Comprehensive overview of the Near-RT RIC implementation by the O-RAN Alliance, including the architecture, internal components and interfaces.	The RIC is implemented in a compute cluster, and ns-3 isused as an O-RAN simulator. The paper provides performance benchmarking of the Near-RT RIC processes such as config delay, subscription config, etc.
[18]	Real-time RIC development	Distributed applications (dApps) that complement xApps and rApps by allowing the implementation of sub-10ms data-driven management and control in real-time at the CU/DU.	Implementation of dApps by leveraging the Colloseum research platform.
[19]	Real-time RIC development	EdgeRIC is a real-time RIC platform that is colocated but decoupled from the RAN nodes, able to access RAN and application-level information and execute AI-optimized actions or policies in real-time (sub-1ms).	Implementation of EdgeRIC on top of the open-source RAN stack srsRAN. The experimental setup uses SDRs and off-the-shelf UEs.

of the O-RAN Alliance and its working, focus, and research groups.

A. WORKING GROUPS

The Alliance has been very active driving standardization to realize the guiding principles of openness, softwarization, and intelligence. Table 7 lists the current work groups in O-RAN focusing on the RU/DU/CU, the RICs the SMO, the interfaces, and interoperability when deploying the stack on a cloud platform (O-Cloud). The principal work to provide *intelligence* in the decision-making and optimization is done in WG2 and WG3. WG2 is concerned with the responsibilities of the Non-RT RIC, all the aspects as presented in Section IV, with work items focusing on ML deployment scenarios and data management (discussed in Section IX).

WG3 is looking to standardize aspects concerning the Near-RT RIC, as presented in Section III, including new E2 service models to enable application use cases defined in Section VIII. WG10 is mostly concerned the inter-dependencies of the SMO and Non-RT RIC, as well as the O1 interface between the SMO to the O-Cloud. For example, the ML deployment options change if there is a separation of the Non-RT RIC and SMO. Finally, WG6 examines cloudification and orchestration, focusing on O1 development as well. In addition, different projects are concerned with cloud-native automation, such as [84], to develop and implement the O1/O2.

B. FOCUS GROUPS

Focus groups in O-RAN concentrate on aspects that transcend the scope of individual workgroups, Table 8. The **Open Source Focus Group (OSFG)** is responsible for taking the specifications developed by the workgroups and implementing it in software that is fully accessible for everyone to use and develop further. The RICs developed by the O-RAN Software Community (OSC) are used by many O-RAN research projects. The **Standards Development Focus Group (SDFG)** plays the leading role on developing the standardization strategies of O-RAN and has the

TABLE 7. O-RAN working groups.

Working Group	Focus Area
WG1	Architecture and use cases
WG2	Non-RT RIC and A1/R1 interfaces
WG3	Near-RT RIC and E2 interface
WG4	Open Fronthaul and interfaces
WG5	Open F1/W1/E1/X2/Xn interfaces
WG6	Cloudification and Orchestration
WG7	White-box hardware
WG8	Stack reference design
WG9	Open X-haul transport
WG10	OAM and O1 interface
WG11	Security

responsibility to relay and align the work with Standard Development Organizations (SDOs) that are relevant for the work standardized in O-RAN, such as ETSI, Linux Foundation, 3GPP, etc.

The **Testing and Integration Focus Group (TIFG)** is responsible of establishing the overall approach for testing and integration for O-RAN, including coordination of test specifications across various WGs. The objective of the group is to provide the end-to-end test & integration specifications, profiles and plans for productization, approaches to meet general requirements, and specifications of processes for performing integration and solution verification. TIFG impacts R&D through the introduction of the Open Testing and Integration Centres (OTIC), public experimental platforms discussed in Section VI, that provide researchers an opportunity go gain experience with vendor software, and that can test alignment to the O-RAN principles in terms of softwarization, openness and interoperability.

Finally, the **Sustainability Focus Group (SuFG)** is responsible to coordinate initiatives within O-RAN that focus on energy efficiency and reduction of environmental impact, in order to create more sustainable mobile networks.

TABLE 8. O-RAN focus groups.

Focus Group	Focus Area
Open Source Focus Group (OSFG)	Open source dev group, establishment of the OSC code base, issues related to it, etc.
Standard Development Focus Group (SDFG)	Standardization strategies
Test and Integration Focus Group (TIFG)	Testing and integration, organization of Plugfests
Sustainability Focus Group (SuFG)	Lead energy efficiency, and sustainability objectives.

C. NEXT GENERATION RESEARCH GROUPS

The O-RAN Alliance has formed the next Generation Research Group (nGRG) to spearhead the development of O-RAN and its evolution in support the 6G requirements, and to align with SDOs such as the International Telecommunication Union Radiocommunication Sector (ITU-R) and 3GPP. Table 9 summarizes the research streams (RS) in nGRG, and their focus areas. The *RIC evolution* is a topic of interest in at least 4 different RSs: **RS02 (architecture), RS03 (AI/ML), RS04 (security), and RS08 (next generation research platforms)**.

TABLE 9. O-RAN next generation research streams.

Next gen research stream	Focus Area
RS01	Requirements
RS02	Architecture
RS03	AI/ML
RS04	Security
RS05	Management
RS06	Cross domain
RS07	Sustainability
RS08	Next generation research platforms

XI. SUMMARY AND CONCLUSION

This paper presented an overview of the O-RAN architecture and specifications, with focus on intelligence aspects that are enabled by virtualization and openness. In the first part we presented the foundational concepts involving the near-RT and non-RT RICs and the critical interfaces to enable xApps and rApps. In the second part of the paper we examined several use cases to illustrate how the various components of the O-RAN architecture interact to provide intelligent end-to-end capabilities.

For the near-RT RIC, we focused on the E2 interface which is critical for the introduction of xApps that operate in 10ms-1s timescales. We discussed the A1 interface between the near-RT RIC and non-RT RIC, which is used to exchange information regarding policies, AI/ML model management, and enrichment information (EI).

For the non-RT RIC side, we focused on the R1 interface between the RIC and the rApps, including data exposure and management aspects. We discussed the O1 interface which is used to collect data from the E2 nodes, and to make configuration changes. The O2 interface - between the non-RT RIC/SMO and the O-Cloud, is used for automated management and provisioning of network functions.

We examine the intelligence aspects of O-RAN by considering the relevant AI/ML workflows. We examined various O-RAN deployment scenarios and the possible placement of various different from the ML pipeline in the near-RT RIC and non-RT RIC/SMO. We also discussed prominent AI/ML approaches that may find application in O-RAN.

Finally, we considered E2E network slice orchestration elucidating the placement of O-RAN components along the orchestration components of the CORE and transport, and the overall E2E Orchestrator. We presented a comprehensive list of the rApp and xApp use cases from the O-RAN Alliance as well as the research community. We discussed research challenges and directions, and presented O-RAN public research platforms. To provide an insight into future O-RAN development we provided an overview of the O-RAN Alliance and its work groups, focus groups, and the next generation research groups.

We hope that this paper has succeeded in providing an accessible means for understanding how the O-RAN Alliance is working to create open, softwarized, intelligent and optimized networks.

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SIMONA MARINOVA received the B.Sc. degree from Ss. Cyril and Methodius University in Skopje, in 2017, and the M.A.Sc. degree from the Department of Electrical and Computer Engineering, University of Toronto, in 2019, where she is currently pursuing the Ph.D. degree. She is a Solution Architect with Bell Canada, where she is involved in strategic planning, architectural development, and the proof of concept validation for various aspects of emerging technologies. Her research interests include O-RAN, end-to-end network slicing and assurance, resource management, and orchestration for future wireless systems.



ALBERTO LEON-GARCIA (Life Fellow, IEEE) was the Founder and the CTO of AcceLight Networks, Ottawa, from 1999 to 2002. AcceLight developed a multi-service switch-router with all-optical fabric. He was the Scientific Director of the NSERC Strategic Network for Smart Applications on Virtual Infrastructures (SAVI) and the Principal Investigator of the Project on Connected Vehicles and Smart Transportation (CVST). SAVI designed and deployed a national testbed in Canada that converges cloud computing and software-defined networking. CVST designed and deployed an application platform for smart transportation. He was the Founder of StreamWorx.ai, which developed massive-scale, real-time streaming analytics software for network operations, and cybersecurity applications. StreamWorx was acquired and is now part of OpenText. He is currently a Professor of electrical and computer engineering with the University of Toronto. He has authored the textbooks: *Probability and Random Processes for Electrical Engineering* and *Communication Networks: Fundamental Concepts and Key Architecture*. His current research interest includes the design of intelligent automated management systems for large-scale infrastructures. He is a fellow of the Royal Society of Canada and a Life Fellow of the Institute of Electronics and Electrical Engineering, for contributions to multiplexing and switching of integrated services traffic. He has held several chair positions with the University of Toronto: the Nortel Chair of Network Architecture and Services, the Skoll Chair of Computer Networks and Innovation, the Canada Research Chair of Autonomic Service Architecture, and a Distinguished Professor of smart infrastructures.